GRADUATION PROJECT / ANALYSIS AND REQUIREMENTS

SmartListing Copilot

Multimodal Listing Quality Control for Fashion E-Commerce

Requirements Documentation

Part I	Functional Requirements Specification (FRS)
Part II	Non-Functional Requirements and Use Cases
Submission date	16 October 2026
Scope authority	SmartListing Copilot – Project Proposal (Revised Scope), 12 October 2026
Objective baseline	Member 1 – Project Planning Document (Revised), OBJ-01 to OBJ-09
Status	Final revised document for academic submission; stakeholder approval pending

Fashion Product Images Small is the sole MVP dataset. Five EDA-selected product categories; Product Type and Color are the only checked attributes. Suggestions are controlled templates, always subject to seller Accept / Edit / Ignore. No external publishing.

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Part I – Functional Requirements Specification (FRS)

Document	Functional Requirements Specification (FRS)
Responsible	Member 5 – Requirements Gathering
Status	Final revised document for academic submission; stakeholder approval pending
Submission date	16 October 2026
Scope authority	SmartListing Copilot – Project Proposal (Revised Scope), 12 October 2026
Objective baseline	Member 1 – Project Planning Document (Revised), OBJ-01 to OBJ-09

> **Scope assurance.** Fashion Product Images Small is the sole MVP dataset. Exactly five product categories will be selected after EDA. Product Type and Color are the only semantic attributes. Suggestions are controlled templates, always subject to seller Accept / Edit / Ignore. No external publishing.

Revision method. Targeted corrections to the existing 25 requirements and eight user stories, with traceability and document-level inspection. Unapproved decisions are explicitly identified, not assumed.

> **Document interpretation.** This specification describes intended behavior and planned tests. A document-level consistency review is not stakeholder validation, a supervisor approval, implementation evidence, or test execution.

1. Document Overview

Purpose. Establish a testable functional specification for the proposed SmartListing Copilot proof of concept, supporting design, implementation planning, integration testing, traceability and academic review.

Coverage. Seller-facing listing intake; parallel computer vision (CV) and natural language processing (NLP); explainable Product Type / Color comparisons; supporting image-quality checks and heuristics; a three-state Review Queue; controlled text suggestions; and seller-controlled Accept / Edit / Ignore decisions.

Audience. Intended operational users are Marketplace Sellers and Catalog Reviewers. Development and academic review contributors use this document as a requirements baseline; no additional in-application role is implied.

Authority and hierarchy. The 12 October 2026 Revised Scope proposal [1] governs scope and routing logic. Member 1 revised planning [2] governs OBJ-01 through OBJ-09 and approved terminology when consistent with [1]. The academic Requirements Gathering Process Guide [4] informs verification, validation and change management. The original Member 5 FRS [3] is the text being revised.

Specification status. Requirements describe a proposed MVP. No completed model training, validation experiment, stakeholder interview, formal walkthrough, requirements approval or system delivery is asserted.

2. Stakeholder Identification

Stakeholder	Classification	Primary need	Expected interaction
Marketplace Seller	Intended primary user	Understand visual/text findings, proposed corrections and approval control	Submit one listing; review findings; Accept / Edit / Ignore
Catalog Reviewer	Intended review user	Consistent severity, explanatory evidence and queue prioritization	Review flagged listings, reason codes and queue states
Development Team	Engineering / document-review stakeholder, not a system role	Bounded scope, stable inputs/outputs and future testability	Implement and test the proposed FastAPI / Streamlit modules
Operations / Support	Indirect beneficiaries only	Reduced ambiguity in catalog quality, if later demonstrated	No separate MVP actor or privileged workflow

Stakeholder	Classification	Primary need	Expected interaction
Academic Supervisor, if applicable	Potential review/approval stakeholder, not an app role	Academic scope and review of unresolved decisions	Future review only; not yet conducted

No administrator account, authorization layer, admin dashboard or operational management feature is defined by the approved MVP. Stakeholder needs are inferred from the documentary scope; no direct interviews or surveys are claimed.

3. Requirements Elicitation

The existing Member 5 elicitation approach is retained. Proposal analysis extracts planned capabilities and constraints from [1]; existing-workflow analysis compares manual catalog review to a proposed first-pass image/text check; scenario-based use-case analysis covers submission, mismatch evidence, review triage and seller decisions; stakeholder-needs analysis relates those scenarios to the expected seller and reviewer experience.

Elicitation evidence classification. Source-derived statements are documentary findings. Seller preferences and reviewer needs are inferred hypotheses until direct stakeholder review occurs. No observations, interviews, meetings, or survey results have been added.

Planned follow-up. Use scenario walkthroughs and the Review Matrix in section 13 to confirm empty-field behavior, conflicting queue flags, terminology, exception handling and usability. Record changes through the controlled log in section 15 before formally baselining the decisions.

4. Functional Requirements

The original 25 FR identifiers, titles, actors, outputs and substantive “The system shall ...” statements are preserved below. Must Have represents the core workflow or a necessary supporting prerequisite; Should Have indicates a simplified supporting capability that may be removed under the proposal fallback. These are proposed specifications and future acceptance cases, not completed tests.

FR-001 | Product Image Upload (Must Have)

Requirement statement	The system shall allow a seller to provide one fashion product image for listing analysis.
Actor / Input	Seller | Product image
Expected output	Image accepted for analysis
Acceptance focus	A supplied image is available to the CV and image-quality branches; a missing image does not start analysis.

FR-002 | Product Title and Description Input (Must Have)

Requirement statement	The system shall accept a product title and a product description as text for attribute-claim extraction.
Actor / Input	Seller | Title; description
Expected output	Text submitted to NLP
Acceptance focus	Submitted title and description are passed to NLP without introducing claims that were not supplied.

FR-003 | Product Category Input (Must Have)

Requirement statement	The system shall accept the seller-provided product category and make it available to the unsupported-category check.
Actor / Input	Seller | Seller-entered category

Expected output	Category available for scope checking
Acceptance focus	The input category is carried to the scope-checking process and compared with supported category signals.

FR-004 | Product Type Classification (Must Have)

Requirement statement	The system shall predict product type within the five EDA-selected categories using the ResNet/MobileNet main CV model, with CLIP zero-shot maintained as baseline and fallback.
Actor / Input	CV module | Uploaded image
Expected output	Predicted type and confidence
Acceptance focus	A supported, assessable image produces a type prediction and confidence; unsupported cases are not forced into a definitive label.

FR-005 | Color Recognition (Must Have)

Requirement statement	The system shall predict product color from the image, with a dominant-color check used to cross-check the visual color decision.
Actor / Input	CV module | Uploaded image
Expected output	Predicted color and confidence
Acceptance focus	An assessable image produces a color prediction and associated confidence; ambiguous outputs can be escalated.

FR-006 | Image Quality Analysis (Must Have)

Requirement statement	The system shall perform basic checks of image resolution, blur and brightness and make relevant quality findings available to review and, when the optional Listing Quality Score is enabled, to scoring.
Actor / Input	CV module | Uploaded image
Expected output	Resolution, blur and brightness findings
Acceptance focus	A blurred image is identified for review with IMAGE_BLUR where the blur rule is triggered; image-quality results feed the score when FR-018 is enabled.

FR-007 | NLP Claim Extraction (Must Have)

Requirement statement	The system shall normalize title and description text and extract only product-type and color claims using a controlled vocabulary, rules and fuzzy matching.
Actor / Input	NLP module | Title; description
Expected output	Claimed type; claimed color
Acceptance focus	Recognized claims are returned in structured type/color fields; unknown attributes are not inferred.

FR-008 | English/Arabic Script Detection (Must Have)

Requirement statement	The system shall identify Arabic characters to select the small fixed Arabic lexicon and otherwise apply the English vocabulary.
Actor / Input	NLP module | Title; description
Expected output	Selected script-based language route
Acceptance focus	Arabic-script input routes to the fixed Arabic lexicon; other input uses the English core.

FR-009 | Image-Text Consistency Checking (Must Have)

Requirement statement	The system shall compare image-supported product type and color with claims extracted from title and description using rules and a lightweight mismatch classifier.
Actor / Input	Comparison engine | CV predictions; NLP claims
Expected output	Attribute comparison results
Acceptance focus	Comparison results identify consistent, contradictory or unclaimed supported attributes without evaluating size, material or price.

FR-010 | Product Type Mismatch Detection (Must Have)

Requirement statement	The system shall flag contradictions between visual product type and text claims and distinguish same-group from different-group contradictions.
Actor / Input	Comparison engine | Predicted type; claimed type
Expected output	Type mismatch finding
Acceptance focus	Different-group type contradictions are Critical; same-group contradictions are Major under the rule table.

FR-011 | Color Mismatch Detection (Must Have)

Requirement statement	The system shall apply color-family and near-shade rules to flag different-family color contradictions while allowing minor or ignored near-shade differences.
Actor / Input	Comparison engine | Predicted color; claimed color
Expected output	Color mismatch finding
Acceptance focus	Different-family contradictions are Major; near shades do not create a Major mismatch solely due to their shade difference.

FR-012 | Missing Attribute Detection (Should Have)

Requirement statement	The system shall identify when an image-supported product type or color is absent from the submitted text rather than contradicted by it.
Actor / Input	Comparison engine | Supported visual type/color; text claims
Expected output	Missing type/color flags
Acceptance focus	An omitted supported color produces MISSING_ATTRIBUTE: Color and review routing; an explicitly contradictory color is treated as a mismatch.

FR-013 | Confidence Scores (Must Have)

Requirement statement	The system shall expose confidence for visual predictions and mismatch findings, distinguishing confidence from verified correctness.
Actor / Input	CV / comparison modules | Model outputs
Expected output	Confidence alongside predictions/warnings
Acceptance focus	A displayed prediction or warning includes its available confidence value without describing it as proof of correctness.

FR-014 | Low-Confidence Handling (Must Have)

Requirement statement	The system shall flag predictions below the configured validation-derived confidence threshold and route affected listings to Needs Review rather than asserting certainty.

Actor / Input	Review engine | Prediction; validation-set threshold
Expected output	Please verify; LOW_CONFIDENCE; queue status
Acceptance focus	A below-threshold prediction receives LOW_CONFIDENCE, a verification message and Needs Review routing.

FR-015 | Unsupported Category Handling (Must Have)

Requirement statement	The system shall flag suspected out-of-scope images when CV and CLIP disagree, the predicted type contradicts the seller category, or CLIP favors the other-product prompt, avoiding a forced definitive type.
Actor / Input	Scope/review engine | Seller category; CV/CLIP signals
Expected output	UNSUPPORTED_CATEGORY; Needs Review
Acceptance focus	A triggered scope check produces UNSUPPORTED_CATEGORY and Needs Review, without presenting a forced definitive classification.

FR-016 | Reason Codes (Must Have)

Requirement statement	The system shall associate applicable findings with the proposal-defined reason codes and a plain-language explanation template.
Actor / Input	Comparison engine | Mismatch and review findings
Expected output	Reason code; explanation
Acceptance focus	Each surfaced warning carries one of TYPE_MISMATCH, COLOR_MISMATCH, LOW_CONFIDENCE, IMAGE_BLUR, MISSING_ATTRIBUTE or UNSUPPORTED_CATEGORY and an explanation.

FR-017 | Mismatch Severity Classification (Must Have)

Requirement statement	The system shall assign severity using the approved product-group and color-family rule table.
Actor / Input	Comparison engine | Attribute mismatch and rule context
Expected output	Critical / Major / Minor-or-ignored severity
Acceptance focus	Cross-group type mismatch is Critical; same-group type and different-family color are Major; near shades are Minor or ignored.

FR-018 | Listing Quality Score (Should Have)

Requirement statement	The system shall calculate a transparent 0–100 Listing Quality Score from image quality, attribute completeness and image-text consistency only.
Actor / Input	Scoring module | Image quality; completeness; consistency
Expected output	Heuristic score from 0 to 100
Acceptance focus	The score uses exactly the three listed components and excludes model confidence and business-return predictions.

FR-019 | Listing Inconsistency Level (Must Have)

Requirement statement	The system shall calculate a heuristic Listing Inconsistency Level from the number and severity of Major and Critical mismatches.
Actor / Input	Scoring module | Major/Critical mismatch findings
Expected output	Low / Medium / High level

Acceptance focus	No Major/Critical mismatch yields Low; one Major yields Medium; two or more Major, or any Critical, yields High.

FR-020 | Review Queue Assignment (Must Have)

Requirement statement	The system shall assign each analyzed listing to the proposal-defined review status based on inconsistency, low confidence, image blur, missing attributes and unsupported-category flags.
Actor / Input	Review engine | Inconsistency level; severity; review flags
Expected output	Ready to publish / Needs review / Critical mismatch
Acceptance focus	A clean low-level listing is Ready to publish; flagged review cases are Needs review; High inconsistency or Critical severity is Critical mismatch.

FR-021 | Template-Based Suggested Fix (Must Have)

Requirement statement	The system shall propose a corrected title and description through controlled templates grounded only in supported type and color predictions, without using an LLM.
Actor / Input	Generation module | Model-supported type/color; original text
Expected output	Suggested corrected title and description
Acceptance focus	Every generated attribute is traceable to predicted type/color and no size, material or price claim is added; suggestion is not auto-applied.

FR-022 | Before/After Comparison (Must Have)

Requirement statement	The system shall present original and proposed listing text side by side and indicate what changed and why.
Actor / Input	Streamlit UI | Original listing; suggested listing; findings
Expected output	Side-by-side before/after display
Acceptance focus	The seller can distinguish original and suggested text and see explanations for changes before deciding.

FR-023 | Accept Suggestion (Must Have)

Requirement statement	The system shall allow the seller to accept a presented correction explicitly without automatically publishing the listing.
Actor / Input	Seller | Displayed suggested correction
Expected output	Seller-approved correction
Acceptance focus	An Accept action records the seller choice in the current review flow and does not perform a marketplace publish action.

FR-024 | Edit Suggestion (Must Have)

Requirement statement	The system shall allow the seller to modify the proposed text before approving their chosen wording, without autonomous publication.
Actor / Input	Seller | Suggested correction; seller edits
Expected output	Seller-edited correction
Acceptance focus	The seller can change suggested wording and choose the edited correction; changes are not automatically published.

FR-025 | Ignore Suggestion (Must Have)

Requirement statement	The system shall allow the seller to ignore a suggested correction and retain the original listing text for that decision.
Actor / Input	Seller | Displayed suggestion
Expected output	Suggestion left unapplied
Acceptance focus	An Ignore action leaves the proposal unapplied and performs no publish action.

5. Functional Requirements Prioritization

MoSCoW	FR identifiers	Authoritative interpretation
Must Have	FR-001–FR-011; FR-013–FR-017; FR-019–FR-025	Core intake, CV/NLP, comparison, severity, explanation, queue and seller control. FR-006 basic resolution / blur / brightness checks remain required; FR-019 stays required for queue routing.
Should Have	FR-012; FR-018	Simplified supporting Missing Attribute Detection (type / color only) and transparent Listing Quality Score (image quality + completeness + consistency only).
Could Have	None of the 25 FRs	Optional CLIP linear probe and simple MLflow logging are implementation/experiment options, not additional operational requirements.
Won't Have	No FR IDs	LLM generation, price checking, return prediction, material/size inference, dynamic pricing, automatic publishing, live marketplace integration, batch processing and cloud production deployment are excluded.

5.1 Priority reconciliation

The Revised Scope distinguishes basic visual-quality checks (part of the core visual pipeline) from the optional three-component Listing Quality Score. Therefore FR-006 remains Must Have, FR-018 remains Should Have and FR-012 remains Should Have. This is an abstraction/dependency clarification, not an expanded MVP. If FR-018 is omitted, FR-006 still reports image-quality findings and review flags; the numeric score/output and dependent score-only test are not claimed. The fallback cut order in [1, §16] is Quality Score, Missing Attribute Detection, MLflow, then Docker; reason codes, Review Queue, suggested fixes and seller control stay.

6. User Stories

The original eight stories and their identifiers are retained; related FRs and acceptance expectations are unchanged.

US-01 — Marketplace Seller

User story	As a Marketplace Seller, I want to submit an image, title, description and category, so that I can receive a listing-quality assessment.
Related FRs	FR-001, FR-002, FR-003
Acceptance	The supplied inputs reach the correct analysis branches.

US-02 — Marketplace Seller

User story	As a Marketplace Seller, I want to see image-based type and color findings, so that I can understand what the product image supports.
Related FRs	FR-004, FR-005, FR-006, FR-013
Acceptance	Predictions, available confidence and quality flags are presented.

US-03 — Marketplace Seller

User story	As a Marketplace Seller, I want to have type and color claims extracted from my text, so that I can identify what my listing actually asserts.
Related FRs	FR-007, FR-008
Acceptance	English and fixed-lexicon Arabic examples produce structured claims.

US-04 — Catalog Reviewer

User story	As a Catalog Reviewer, I want to inspect mismatches and their reasons, so that I can triage incorrect listings consistently.
Related FRs	FR-009, FR-010, FR-011, FR-016, FR-017
Acceptance	Rule-consistent codes, explanations and severity are shown.

US-05 — Catalog Reviewer

User story	As a Catalog Reviewer, I want to see missing attributes and uncertain or unsupported products, so that I can avoid accepting ambiguous listings.
Related FRs	FR-012, FR-014, FR-015
Acceptance	Missing attributes and uncertain/suspected unsupported items receive review flags.

US-06 — Catalog Reviewer

User story	As a Catalog Reviewer, I want to see scores, inconsistency levels and queue statuses, so that I can prioritize human attention.
Related FRs	FR-018, FR-019, FR-020
Acceptance	Scoring uses stated heuristic signals and queue routing follows the proposal.

US-07 — Marketplace Seller

User story	As a Marketplace Seller, I want to compare a proposed listing with my original, so that I can understand suggested corrections before acting.
Related FRs	FR-021, FR-022
Acceptance	Only supported type/color appears in the suggestion and changes are visible.

US-08 — Marketplace Seller

User story	As a Marketplace Seller, I want to accept, edit or ignore proposed corrections, so that I can retain final control over wording.
Related FRs	FR-023, FR-024, FR-025
Acceptance	Each seller choice is respected and no listing is automatically published.

7. Input Requirements

A required field means that the proposed listing intake identifies that input; it does not itself decide whether an empty value is permitted. Unspecified error messages, encodings and numeric limits are not inferred. The following table reconciles FR-001 through FR-003 and the original section 7 with [1].

Input	Required status	Validation rule	Invalid / missing input behavior	Confirmation status

Input	Required status	Validation rule	Invalid / missing input behavior	Confirmation status
Product Image	Required for analysis	Single readable image; processable image must reach CV and image-quality branches. No accepted encodings or size limit specified.	Missing: analysis does not start (confirmed). Unreadable / corrupt / unsupported encoding: normal analysis cannot proceed; error presentation not specified.	Missing behavior confirmed; error response, encoding limits To Be Confirmed
Product Title	Input specified; empty-value rule open	Pass supplied text for type/color claims; do not infer unsupported claims. Dataset proxy: productDisplayName.	Blank/missing title: reject or treat as absent claims not specified; do not assume silent acceptance.	To Be Confirmed – Team/Supervisor Decision Required
Product Description	Input specified; empty-value rule open	Rule-based English / bounded Arabic extraction; evaluation descriptions are manually written or synthetic, not a dataset column.	Blank/missing description: rejection vs missing textual claims remains undecided.	To Be Confirmed – Team/Supervisor Decision Required
Product Category	Required as seller-provided category	Pass value to scope comparison with EDA-selected supported types and visual signals.	Out-of-scope product triggers UNSUPPORTED_CATEGORY and Needs review (if detected); blank/malformed label handling not defined.	Unsupported-category workflow confirmed; blank/invalid input To Be Confirmed

7.1 Validation decision register

ID	Decision required
IN-01	Accept or reject empty Product Title; identify the user-facing response and AC-002 fixture.
IN-02	Accept or reject empty Product Description; if accepted, determine whether it contributes missing claims or only an empty NLP result and adjust FR-012/AC-012 as needed.
IN-03	Specify how unreadable, corrupted or unsupported-encoding image files are detected and reported under FR-001/AC-001 without inventing extensions, size caps or thresholds.
IN-04	Define missing/malformed category validation separately from the existing FR-015 detection of a suspected out-of-scope product.

All four are pending team/supervisor review.

8. Output Requirements

Output	Behavior / interpretation
Predicted Product Type	Supported EDA-selected type when assessable, or review-required indication; no forced category for suspected unsupported products.
Predicted Color	Image-supported color label with available confidence; color-family ambiguity is respected.
Confidence Scores	Calibrated CV prediction confidence and available warning/mismatch confidence; never guarantees correctness.
Extracted Text Claims	Normalized type and color claims and script-based lexicon route; no unsupported attributes.
Mismatch Report	Visual-versus-text evidence, contradiction/omission as applicable, severity and plain-language reasoning.
Reason Codes	TYPE_MISMATCH, COLOR_MISMATCH, LOW_CONFIDENCE, IMAGE_BLUR, MISSING_ATTRIBUTE, UNSUPPORTED_CATEGORY.
Severity Level	Critical / Major / Minor-or-ignored under the authoritative type-group/color-family rules.
Listing Quality Score	Optional supporting 0–100 three-component heuristic only if FR-018 is retained; not a business-risk prediction.

Output	Behavior / interpretation
Listing Inconsistency Level	Low / Medium / High based only on Major/Critical mismatches, not return probability.
Review Queue Status	Ready to publish / Needs review / Critical mismatch; local decision-support label only. Medium-only and combined-flag precedence still open.
Suggested Corrected Listing	Template-controlled title/description from model-supported Product Type and Color, not LLM output.
Before/After Comparison	Original versus suggested text with changed wording and reasons, followed by seller Accept / Edit / Ignore.

9. Functional Workflow

The original two-branch workflow is retained. The seller provides image, title, description and category through Streamlit. FastAPI orchestrates parallel image and text processing. The CV branch supplies visual attributes/confidence and quality signals; the NLP branch supplies structured claims; the comparison engine derives mismatches, explanations and supporting heuristics. The queue and controlled suggestion lead to a seller decision. None of these stages performs external marketplace publishing.

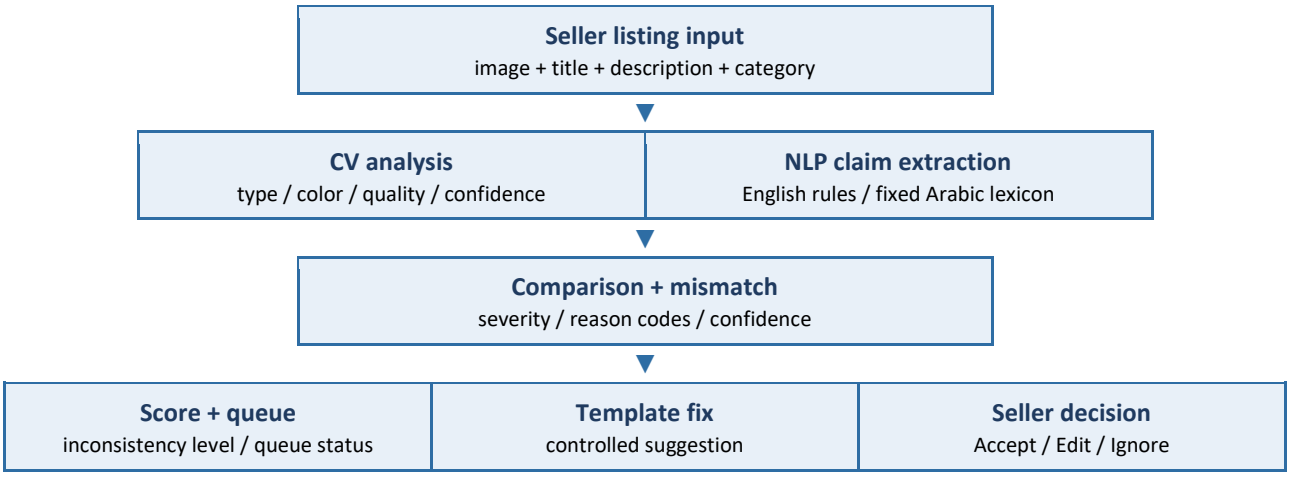


Figure 2. Functional sequence. The CV and NLP branches operate in parallel; the final choice is under human control. This is a proposed flow, not an implemented integration.

10. Functional Acceptance Criteria

AC-001 to AC-025 remain one-to-one with FR-001 to FR-025. These are future fixture-driven acceptance tests, not executed test results. Values requiring validation-set tuning or team decisions must be frozen before their related tests can be signed off.

ID	FR	Given / When / Then
AC-001	FR-001	Given a valid image and the listing intake view; when the seller submits the image; then it is available to the CV branch and a missing image prevents analysis.
AC-002	FR-002	Given a title and a description; when the seller submits the text; then both fields reach NLP unchanged in meaning.
AC-003	FR-003	Given a seller category; when listing analysis starts; then the category is available to the unsupported-category comparison.
AC-004	FR-004	Given an assessable image of a selected fashion type; when the main CV model is run, or CLIP fallback is used; then a supported type and confidence are produced unless a scope check requires review.
AC-005	FR-005	Given an assessable image with an evaluable color; when visual color analysis and dominant-color cross-check run; then a color and confidence are reported, with ambiguous outcomes eligible for review.

ID	FR	Given / When / Then
AC-006	FR-006	Given a deliberately blurred image; when the image-quality checks run; then the blur finding is flagged for review and made available to the quality score if FR-018 is retained; blur review stays required.
AC-007	FR-007	Given text containing a recognized color and type; when controlled NLP extraction runs; then structured claims identify those two attributes and introduce no others.
AC-008	FR-008	Given Arabic-script text and English-only text in separate cases; when script routing runs; then Arabic-script text selects the fixed lexicon and the English-only case uses English rules.
AC-009	FR-009	Given defined CV predictions and extracted NLP claims; when attribute comparison runs; then type/color consistency findings are returned using rules and lightweight classifier signals.
AC-010	FR-010	Given a cross-group and a same-group type contradiction; when the type mismatch rules run; then the first is Critical and the second Major.
AC-011	FR-011	Given different-family and near-shade color pairs; when color rules run; then the different-family case is Major and the near shade is Minor or ignored.
AC-012	FR-012	Given a visible supported color omitted from text; when missing-attribute analysis runs; then MISSING_ATTRIBUTE: Color appears without falsely reporting a color contradiction.
AC-013	FR-013	Given predictions and generated warnings; when results are rendered; then the available confidence accompanies the findings and is not represented as correctness.
AC-014	FR-014	Given a prediction below the validation-set threshold; when confidence checks run; then LOW_CONFIDENCE, Please verify and Needs review are emitted.
AC-015	FR-015	Given CV/CLIP disagreement, a seller-category contradiction or other-product preference; when unsupported-category checks run; then UNSUPPORTED_CATEGORY is raised, forced definitive type is avoided and status is Needs review.
AC-016	FR-016	Given each supported warning class; when findings are displayed; then the corresponding approved reason code and readable explanation appear.
AC-017	FR-017	Given type and color mismatch fixtures covering approved severity cases; when severity is determined; then Critical/Major/Minor-or-ignored labels follow the proposal rule table.
AC-018	FR-018	Given fixed image-quality, completeness and consistency inputs; when scoring runs; then the output lies in 0–100 and uses only those three components, excluding confidence.
AC-019	FR-019	Given fixtures with zero, one and two Major mismatches and one Critical mismatch; when inconsistency is calculated; then outcomes are Low, Medium, High and High, respectively.
AC-020	FR-020	Given clean, low-confidence and Critical fixtures; when review routing runs; then statuses are Ready to publish, Needs review and Critical mismatch, respectively. This fixture set does not establish a queue outcome for a lone Major mismatch (Medium level) or precedence among conflicting flags; acceptance of those cases is blocked pending documented decision Q-01/Q-02.
AC-021	FR-021	Given model-supported type/color and original listing text; when a template suggestion is generated; then only supported type/color are used and no LLM-derived attributes or auto-application occur.
AC-022	FR-022	Given an original listing and a generated suggestion; when the seller opens the comparison; then before/after text and the rationale for changes are visible.
AC-023	FR-023	Given a proposed correction; when the seller selects Accept; then the correction is seller-approved within the review flow and not externally published.
AC-024	FR-024	Given an editable suggested correction; when the seller changes text and confirms the edit; then the seller-selected revised wording is retained for that decision without publication.
AC-025	FR-025	Given an unapplied suggestion; when the seller selects Ignore; then original listing text remains unchanged and nothing is published.

11. Requirements Dependencies

Upstream FRs	Dependent FRs	Functional dependency
FR-001, FR-003	FR-004–FR-006, FR-015	Image and category are prerequisites for CV, quality and out-of-scope checks.
FR-002	FR-007, FR-008	Submitted title / description feed claim extraction and script detection.
FR-004, FR-005, FR-007, FR-008	FR-009–FR-012	Comparison and missing-attribute checks depend on aligned visual and textual claims.
FR-004, FR-005, FR-009–FR-011	FR-013, FR-014, FR-016, FR-017	Model evidence and comparison signals support confidence, explanations and severity.
FR-006, FR-009, FR-012 (if retained)	FR-018 (if retained)	Optional score combines quality, completeness and consistency; optional supporting capability dependencies are conditional.
FR-010, FR-011, FR-017	FR-019	Major/Critical counts yield the Inconsistency Level.
FR-006, FR-012 (if retained), FR-014–FR-017, FR-019	FR-020	Queue status depends on inconsistency and supported review triggers; unresolved precedence documented in section 12.
FR-004, FR-005, FR-014, FR-015	FR-021	Suggestions must use only supported attributes and respect uncertainty/out-of-scope review.
FR-021	FR-022–FR-025	Seller compares and chooses how to treat the proposed text.

> **Integration boundary.** The proposal specifies a unified Pydantic contract, FastAPI orchestration and Streamlit presentation. These are implementation boundaries, not additional user-facing requirements or new Member 6 use cases.

12. Assumptions, Constraints and Open Decisions

12.1 Fixed scope constraints

Fashion Product Images Small exclusively for MVP training, validation, testing and pipeline evaluation; five final categories selected after EDA; Product Type and Color only; ResNet/MobileNet supervised transfer learning with CLIP zero-shot baseline/fallback; rules and lightweight mismatch classifier; English core with fixed Arabic lexicon; FastAPI/Streamlit; controlled templates; seller Accept / Edit / Ignore. Evaluation descriptions are manually authored or synthetic and clearly labeled. No external publishing or other dataset is permitted.

12.2 FR-020 review-routing decision table

The source [1, §§8.3, 8.4, 12.2] defines the Inconsistency Level and three queue rules but omits certain precedence combinations. The table reports only specified results as approved source behavior. “To Be Confirmed – Team/Supervisor Decision Required” denotes a gap, not an implied feature or accepted default.

Scenario	Severity / Flag	Inconsistency Level	Review Queue	Decision Status
Clean listing, no review flags	No Major/Critical; none	Low	Ready to publish	Specified [1]
Exactly one Major contradiction, no other flags	Major	Medium	To Be Confirmed – Team/Supervisor Decision Required	Open Q-01
Two or more Major contradictions	Major × 2+	High	Critical mismatch	Specified [1]

Scenario	Severity / Flag	Inconsistency Level	Review Queue	Decision Status
At least one Critical contradiction	Critical	High	Critical mismatch	Specified [1]
Low-confidence prediction only	LOW_CONFIDENCE review flag	Low (flag does not alter level)	Needs review	Specified [1]
Blurred image only	IMAGE_BLUR review flag	Low (flag does not alter level)	Needs review	Specified [1]
Missing supported Type / Color claim only	MISSING_ATTRIBUTE (if FR-012 retained)	Low (no Major/Critical)	Needs review	Specified [1]; supporting feature
Suspected unsupported category only	UNSUPPORTED_CATEGORY	Low if no Major/Critical	Needs review; avoid forced type	Specified [1]
Multiple flags with conflicting queue triggers	Mixed, e.g. Critical + low confidence	High if Critical; otherwise defined by Major count	To Be Confirmed – Team/Supervisor Decision Required	Open Q-02: precedence
Near-shade difference, treated as Minor	Minor / COLOR_MISMATCH	Low	Needs review	Specified for retained Minor flag [1]
Near-shade difference, ignored under color rule	Ignored: no active mismatch	Low	Ready to publish only if no other flags	Conditional result from [1]

Interpretation. A low-confidence or blurred image alone does not increase the Inconsistency Level. Missing-attribute detection is optional, but if retained its flag routes to Needs review. A Minor near-shade mismatch routes for review when preserved as a finding; if the shade difference is ignored by the approved tolerance and no other flag exists, the case meets clean/Low criteria. The rule for a Medium-only listing is not specified. Simultaneous high-severity and review-only flags require explicit precedence before a single deterministic status can be tested.

12.3 Resolution workflow and controlled changes

Q-01: Select and document the queue state for exactly one Major mismatch with Medium Inconsistency. Recommendation for discussion only (NOT APPROVED): consider Needs review to avoid a false “Ready” status.

Q-02: Confirm precedence for simultaneous High/Critical and review-only flags, and whether accumulated flags have any additional tie-breaking logic. Recommendation for discussion only (NOT APPROVED): consider prioritizing Critical mismatch when High/Critical is present.

Neither proposal is implemented as an accepted rule until a team/supervisor decision is recorded. On approval, the normative FR-020 rule text or accompanying routing table is changed, AC-020 is extended with explicit Medium-only and combined-flag fixtures, sections 8 and 11 and the traceability review status are updated, and the decision/evidence/date is entered in section 15. Until then AC-020 is only partially specified for routing edge cases.

12.4 Additional unresolved configuration

Pending: final five EDA-based articleType classes; class-group boundaries and color-family / near-shade tolerance; confidence threshold selected on validation data before held-out testing; exact optional score weights; empty-title and empty-description decisions; image codec/error-response rules; malformed category handling; and the duration/representation of seller decisions within the proof of concept. No numeric thresholds, accounts or audit-trail storage are implied.

13. Requirements Verification and Validation Plan

13.1 Definitions and evidence status

Requirements Verification examines the specification artifact for correctness against source documents, completeness, internal consistency, feasibility, testability, traceability and scope alignment. Requirements

Validation asks whether those written requirements accurately reflect the needs and expectations of the real intended sellers, reviewers and authorized project stakeholders. A cross-document inspection can substantiate verification findings but cannot substitute for direct stakeholder validation.

Methods. Completed within this document revision: Document Inspection, Cross-Document Consistency Review, traceability/ID inspection and Acceptance Criteria Inspection. Proposed but NOT yet conducted: Scenario-Based Walkthrough with stakeholders, stakeholder confirmation of edge-case decisions and formal sign-off. Methodology follows [4, steps 5–7]; no meeting, survey, reviewer identity or approval is invented.

13.2 Review criteria and coverage

Review criterion	Inspection question / evidence
Correctness	Does the FR reflect the Revised Scope rather than an unsupported new behavior?
Completeness	Are normal flow, exception cases, input/output needs and dependencies documented, or explicitly left open?
Consistency	Are FRs, ACs, queue/severity rules, MoSCoW tiers and Member 1 objectives mutually compatible?
Feasibility	Does the FR fit the five-class, two-attribute, single-dataset prototype and planned technology stack?
Testability	Does each FR have a prospective fixture-based AC with observable output; are unresolved test decisions tracked?
Traceability	Is each FR uniquely mapped to a valid project objective, user story and acceptance-test ID?
Scope Alignment	Is the core/supporting boundary intact and are prohibited capabilities absent?

13.3 Requirements Review Matrix

Requirement ID	Review Criteria	Validation Method	Review Status	Open Issue / Required Action
FR-001	Correctness; completeness; testability	Document Inspection; AC Inspection	Document issue OPEN; Pending Stakeholder Validation	IN-03: unreadable/codec error response pending.
FR-002	Completeness; testability	Document Inspection; Scenario Walkthrough (planned)	Document issue OPEN; Pending Stakeholder Validation	IN-01/IN-02: empty title/description rules pending.
FR-003	Completeness; consistency	Cross-Document Review; Scenario Walkthrough (planned)	Document issue OPEN; Pending Stakeholder Validation	IN-04: blank/malformed category behavior pending.
FR-004	Correctness; scope	Cross-Document Review; AC Inspection	Document check complete; Pending Stakeholder Validation	Final EDA-selected classes and thresholds pending.
FR-005	Correctness; testability	Cross-Document Review; AC Inspection	Document check complete; Pending Stakeholder Validation	Color alias/family tolerance after EDA pending.
FR-006	Priority; consistency; scope	Scope Review; AC Inspection	Document check complete; Pending Stakeholder Validation	Blur thresholds pending; numeric score is conditional.
FR-007	Correctness; scope	Document Inspection; AC Inspection	Document check complete; Pending Stakeholder Validation	No unresolved structural issue; limited lexicon remains.
FR-008	Feasibility; testability	Document Inspection; AC Inspection	Document check complete; Pending Stakeholder Validation	Arabic lexicon examples require future fixture tests.

Requirement ID	Review Criteria	Validation Method	Review Status	Open Issue / Required Action
FR-009	Consistency; correctness	Scope Review; AC Inspection	Document check complete; Pending Stakeholder Validation	Mismatch-classifier contribution untested; rules remain authoritative.
FR-010	Correctness; testability	Rule Review; AC Inspection	Document issue OPEN; Pending Stakeholder Validation	Class group membership needs EDA confirmation.
FR-011	Correctness; testability	Rule Review; AC Inspection	Document issue OPEN; Pending Stakeholder Validation	Near-shade tolerance / ignored-vs-Minor selection pending.
FR-012	Priority; scope	Scope Review; AC Inspection	Document check complete; Pending Stakeholder Validation	Supporting feature may be cut under fallback.
FR-013	Testability; correctness	Evidence Review; AC Inspection	Document check complete; Pending Stakeholder Validation	Confidence calibration not yet tested.
FR-014	Feasibility; testability	Scenario Walkthrough (planned); AC Inspection	Document issue OPEN; Pending Stakeholder Validation	Validation-derived confidence threshold pending.
FR-015	Correctness; scope	Scenario Walkthrough (planned); Scope Review	Document check complete; Pending Stakeholder Validation	Final five classes and scope-trigger behavior require future tests.
FR-016	Completeness; traceability	Document Inspection; AC Inspection	Document check complete; Pending Stakeholder Validation	Reason-code templates require stakeholder comprehension check.
FR-017	Consistency; testability	Rule Review; AC Inspection	Document issue OPEN; Pending Stakeholder Validation	Group and family rule parameters require decision.
FR-018	Priority; testability	Priority Review; AC Inspection	Document issue OPEN; Pending Stakeholder Validation	Score weights pending; optional under fallback.
FR-019	Correctness; testability	Cross-Document Review; AC Inspection	Document check complete; Pending Stakeholder Validation	Cases defined by proposal; future fixtures not run.
FR-020	Consistency; testability; completeness	Cross-Document Review; Scenario Walkthrough (planned)	Document issue OPEN; Pending Stakeholder Validation	Q-01 Medium-only; Q-02 combined-flag precedence open.
FR-021	Scope; correctness	Scope Review; AC Inspection	Document check complete; Pending Stakeholder Validation	Review safe behavior when image/type uncertainty occurs.
FR-022	Testability; completeness	Scenario Walkthrough (planned); AC Inspection	Document check complete; Pending Stakeholder Validation	Future seller walkthrough of explanation clarity.
FR-023	Correctness; scope	AC Inspection; Stakeholder Review (planned)	Document check complete; Pending Stakeholder Validation	Future seller confirmation; no publish action approved.

Requirement ID	Review Criteria	Validation Method	Review Status	Open Issue / Required Action
FR-024	Correctness; scope	AC Inspection; Stakeholder Review (planned)	Document issue OPEN; Pending Stakeholder Validation	Future seller confirmation; edit persistence details open.
FR-025	Correctness; scope	AC Inspection; Stakeholder Review (planned)	Document check complete; Pending Stakeholder Validation	Future seller confirmation; no publish action approved.

13.4 Review and Sign-off Record

Reviewer	Pending – no named reviewer evidenced
Role	Pending – team/supervisor reviewer to be designated
Review Date	Not Yet Conducted (formal review)
Review Scope	Proposed: scope, 25 FRs, 8 USs, 25 ACs, input/queue exceptions, traceability and priorities
Decision	Pending Stakeholder Validation; no approval or rejection recorded
Open Issues	Q-01, Q-02; IN-01 to IN-04; EDA-derived groups/thresholds; optional score weights; seller workflow details

Document-level inspection during revision was completed by the authoring review process without recording an independent human reviewer. It is not a signed acceptance or supervisor confirmation. The formal sign-off record above intentionally remains pending.

14. Requirements Traceability

Objective IDs and descriptions reproduce the Member 1 revised planning baseline [2, §4]. Mappings are behavior-based: intake, visual inference, text extraction, contradiction analysis, explanations, review prioritization, controlled suggested text, and seller decisions are distinguished. OBJ-09 relates to future evaluation deliverables rather than a new user-facing feature and is therefore not artificially attached to an FR.

Member 1 objective	Exact objective description
OBJ-01	Establish the end-to-end image + text listing-checking pipeline
OBJ-02	Predict visual Product Type and Color for the selected five classes
OBJ-03	Extract Product Type and Color claims from title / description text
OBJ-04	Detect discrepancies between CV outputs and NLP claims
OBJ-05	Attach understandable evidence and reason codes to every warning
OBJ-06	Prioritize listings for human review
OBJ-07	Generate bounded suggested title / description corrections
OBJ-08	Ensure that suggested edits remain under seller control
OBJ-09	Evaluate the pipeline with transparent evidence and limitations

14.1 Requirements Traceability Matrix

FR ID	Requirement Name	Related Project Objective	Related User Story	Acceptance Test ID
FR-001	Product Image Upload	OBJ-01	US-01	AC-001
FR-002	Product Title and Description Input	OBJ-01	US-01	AC-002

FR ID	Requirement Name	Related Project Objective	Related User Story	Acceptance Test ID
FR-003	Product Category Input	OBJ-01	US-01	AC-003
FR-004	Product Type Classification	OBJ-02	US-02	AC-004
FR-005	Color Recognition	OBJ-02	US-02	AC-005
FR-006	Image Quality Analysis	OBJ-01, OBJ-06	US-02	AC-006
FR-007	NLP Claim Extraction	OBJ-03	US-03	AC-007
FR-008	English/Arabic Script Detection	OBJ-03	US-03	AC-008
FR-009	Image-Text Consistency Checking	OBJ-04	US-04	AC-009
FR-010	Product Type Mismatch Detection	OBJ-04	US-04	AC-010
FR-011	Color Mismatch Detection	OBJ-04	US-04	AC-011
FR-012	Missing Attribute Detection	OBJ-06	US-05	AC-012
FR-013	Confidence Scores	OBJ-02, OBJ-05	US-02	AC-013
FR-014	Low-Confidence Handling	OBJ-05, OBJ-06	US-05	AC-014
FR-015	Unsupported Category Handling	OBJ-02, OBJ-06	US-05	AC-015
FR-016	Reason Codes	OBJ-05	US-04	AC-016
FR-017	Mismatch Severity Classification	OBJ-04, OBJ-05	US-04	AC-017
FR-018	Listing Quality Score	OBJ-01	US-06	AC-018
FR-019	Listing Inconsistency Level	OBJ-06	US-06	AC-019
FR-020	Review Queue Assignment	OBJ-06	US-06	AC-020
FR-021	Template-Based Suggested Fix	OBJ-07	US-07	AC-021
FR-022	Before/After Comparison	OBJ-07	US-07	AC-022
FR-023	Accept Suggestion	OBJ-08	US-08	AC-023
FR-024	Edit Suggestion	OBJ-08	US-08	AC-024
FR-025	Ignore Suggestion	OBJ-08	US-08	AC-025

14.2 Mapping rationale and evaluation boundary

FR-004/005 map to visual objective OBJ-02, while FR-007/008 map to NLP objective OBJ-03. FR-009–011/017 implement contradiction rules (OBJ-04); FR-013/016 and explanatory severity support OBJ-05. FR-012/014/015/019/020 concern review signals or decisions (OBJ-06). FR-021/022 concern template suggestions and comparison (OBJ-07); FR-023–025 concern seller control (OBJ-08). FR-006 is a pipeline-quality prerequisite with review implications (OBJ-01/06), while supporting FR-018 is an interpretable pipeline score (OBJ-01), not an OBJ-09 evaluation experiment.

OBJ-09 is represented in Member 1's planned evaluation protocol and reporting, not by inventing a new functional requirement. Model evaluation, synthetic-vs-manual sample distinction, accuracy/F1 and thresholds belong to later evaluation work; acceptance cases here provide traceability for eventual verification, not reports of execution.

15. Requirements Change Log

Change ID	Section	Previous Issue	Revision Made	Reason
CH-01	14	Legacy internal OBJ-1 to OBJ-5 mapping	Replaced with Member 1 OBJ-01 to OBJ-09 nomenclature; traceability re-mapped by behavior.	Cross-document objective consistency.

Change ID	Section	Previous Issue	Revision Made	Reason
CH-02	14	Traceability rows lacked names and valid approved objective IDs.	Added all 25 FR names, one valid US and one AC per row; multiple OBJs only where justified.	Auditable requirements coverage.
CH-03	13	No dedicated verification / stakeholder-validation procedure.	Added criteria, 25-row review matrix, methods, evidence status, sign-off record.	Follow academic methodology; avoid fictional approvals.
CH-04	12, 10	FR-020 medium-only queue and conflicting flag precedence unclear.	Added route scenarios and open Q-01/Q-02; left approved queue cases intact; annotated AC-020.	No unauthorized routing default.
CH-05	7, 12	Input validation exceptions not specified.	Added input-rule/status table and open IN-01–IN-04.	Prevent invented empty-field, codec or invalid-input behavior.
CH-06	4, 5, 11	Core image checks could be mistaken for optional Score.	Kept FR-006 Must and FR-018 Should; made score-only dependency conditional.	Preserve required blur/quality review while honoring fallback.
CH-07	6, 10	Need to preserve original identifiers and tests.	Kept FR-001–025, US-01–08, AC-001–025; targeted language updates only where needed.	Maintain stable Member 6 integration baseline.
CH-08	8, 9, 16	Explanatory and reference consistency.	Clarified conditional outputs, retained workflow design and added source hierarchy / revised references.	Improve academic document auditability.

Change-log status. These are document revisions only. No team or supervisor approval of the open requirements decisions is implied; any future baseline change requires evidence, reviewer, date and corresponding updates to FR/AC/traceability.

16. References

Ref.	Reference
[1]	SmartListing Copilot. SmartListing Copilot – Project Proposal (Revised Scope: Fashion Product Images Small). 12 October 2026. 22 pages. Primary scope authority.
[2]	SmartListing Copilot. Member 1 – Project Planning Document (Revised). Submission baseline 16 October 2026. 12 pages. Objective IDs OBJ-01 to OBJ-09 and planning terminology.
[3]	SmartListing Copilot. SmartListing_Copilot_Functional_Requirements.pdf. Original Member 5 Functional Requirements Specification. 16 pages. Text and identifier preservation baseline.
[4]	Requirements Gathering Process Guide. Two-page academic methodology handout provided with project materials. Stakeholder identification, elicitation, prioritization, verification/validation and change control.

Reference note. The above titles, dates, roles and page counts are taken from the supplied or recovered project documents. No unpublished interview source, external experiment, supervisor endorsement or unverified research citation has been introduced.

Part II – Non-Functional Requirements and Use Cases

0. Document Basis and Conventions

Evidence discipline. Both source documents avoid claiming tested results, and this document follows them. No performance value is measured, and no stakeholder has approved anything. Where the sources fix no number, the requirement is stated as "threshold TBD (frozen on validation data or by team decision)". No numbers have been invented.

Convention	Meaning
NFR-xxx-nn	Non-functional requirement ID (PERF, SEC, REL, USA, MNT, CON, TEC)
UC-nn	Use case ID
TAC-nn	Non-functional / use-case acceptance criterion (kept separate from AC-001 to AC-025)
Confirmed	Prescribed by the Revised Scope or the Member 1 / Member 5 documents
Proposed	Engineering expectation derived from the design, not a cited regulation
TBD	Open decision; no default is assumed
Priority	Must / Should, using the same MoSCoW logic as the FRS

Scope reminders carried into every section. Fashion Product Images Small is the only dataset. There are five EDA-selected categories. Only Product Type and Color are checked. There is no auto-publishing, no LLM generation, no batch CSV, no cloud production deployment and no admin role.

1. Performance Requirements

OBJ-09 and SC-10 require that response time and catch rate are measured. No numeric targets are set before baseline experiments, so the requirements below fix what must be measured and how, and leave the numbers to be frozen.

ID	Requirement	Priority	Basis	Verification
NFR-PERF-01	The system shall measure end-to-end response time from listing submission to displayed findings, reported separately for the CV branch, the NLP branch, comparison and queue assignment.	Must	SC-10, OBJ-09	Timing test over the manual evaluation set
NFR-PERF-02	The CV and NLP branches shall run in parallel, so that total latency is bounded by the slower branch plus comparison, not by the sum of both.	Must	FRS §9, Planning §9	Timing comparison: parallel vs sequential execution
NFR-PERF-03	The response-time target shall be set after the first prototype timing and recorded before final evaluation. Threshold TBD.	Must	Proposal discipline	Documented decision with date
NFR-PERF-04	Model accuracy shall be reported per attribute (accuracy, macro-F1, confusion matrix, ECE for Product Type and Color). Thresholds are frozen on validation data after baselines and before held-out testing.	Must	OBJ-02, SC-02	Evaluation report
NFR-PERF-05	Mismatch detection quality shall be reported as precision, recall, F1 and false-alarm rate for wrong-type, wrong-color, combined-error and correct-listing cases.	Must	OBJ-04, SC-03	Evaluation report
NFR-PERF-06	NLP claim extraction shall be reported as precision, recall and F1 by claim type and by language (English / Arabic).	Must	OBJ-03	Evaluation report

ID	Requirement	Priority	Basis	Verification
NFR-PERF-07	The end-to-end catch rate shall be reported on synthetic and manual sets separately, never combined.	Must	SC-09, SC-10	Report inspection
NFR-PERF-08	Model weights shall be loaded once at service startup and reused across requests, not reloaded per request.	Should	Proposed (FastAPI design)	Code inspection and repeated-request timing
NFR-PERF-09	The system shall process one listing per request. Throughput and concurrency targets are not set because batch processing is out of scope.	Must	Out-of-scope list	Scope review

> **Example (NFR-PERF-01).** For one submission, the timing log records: image preprocessing 0.00 s | CV inference 0.00 s | NLP extraction 0.00 s | comparison 0.00 s | queue + template 0.00 s | total 0.00 s. (The zeros are placeholders; real values come from the testing phase.) The report gives the mean and spread over the 100–150 manual listings.

2. Security Requirements

The MVP has no accounts, no administrator role, no database of users, no live marketplace link and no cloud production deployment. Security requirements are therefore scoped to what a local academic prototype actually exposes, and no legal or industry compliance is claimed (consistent with the Planning Document §3).

ID	Requirement	Priority	Basis	Verification
NFR-SEC-01	No autonomous publishing. No code path shall send a listing to an external marketplace, and Accept / Edit / Ignore shall never trigger a publish action.	Must	FR-023–025, OBJ-08	Interaction tests and code inspection (AC-023–025)
NFR-SEC-02	Image upload validation. The system shall accept only a single readable image. Accepted encodings and size limits are TBD (IN-03). The system shall not execute or interpret uploaded content as code.	Must	FR-001, IN-03	Upload tests with valid, corrupt and wrong-type files
NFR-SEC-03	Input validation on text and category. Title, description and category shall be treated strictly as data, validated through the shared Pydantic contract, and never evaluated or executed.	Must	Pydantic contract, FR-002/003	Malformed and special-character input tests
NFR-SEC-04	Safe error responses. Errors shall not expose stack traces, file paths or model internals to the user.	Should	Proposed	Fault-injection test
NFR-SEC-05	Data provenance. Only dataset images and metadata from Fashion Product Images Small shall be used, and synthetic or manual descriptions shall be labeled by origin.	Must	Planning §8, FRS §12.1	Data-card inspection
NFR-SEC-06	Data minimization. The prototype shall not require personal data. Uploaded images and text shall not be sent to external services, and CLIP and CV inference shall run locally.	Should	Proposed	Dependency and network-call review
NFR-SEC-07	Local-only deployment. The prototype runs on the team's local environment for demonstration, with no public endpoint.	Must	Out-of-scope: production cloud	Deployment review
NFR-SEC-08	Dependency hygiene. Dependencies shall be pinned in the requirements file, and model weights shall come from official sources.	Should	Proposed	Requirements-file inspection
NFR-SEC-09	No authentication or authorization layer is defined, and none shall be implied. If one is added later, it requires a controlled change (FRS §15).	Must	FRS §2	Scope review

3. Reliability Requirements

ID	Requirement	Priority	Basis	Verification
NFR-REL-01	Graceful degradation under uncertainty. Low confidence, blur or unsupported categories shall route to Needs review and shall never be silently accepted as Ready to publish.	Must	FR-014, FR-015, SC-08	AC-006, AC-014, AC-015
NFR-REL-02	Fallback model. If the primary ResNet/MobileNet model is unavailable or disagrees with CLIP beyond the scope-check rule, the system shall use CLIP zero-shot or escalate to review.	Must	FR-004, FR-015	Disable-primary-model test
NFR-REL-03	Deterministic decision logic. Given identical model outputs and claims, severity, inconsistency level, queue status and suggestion shall be identical every time. Model inference stays fixed (eval mode, no sampling).	Must	FR-017, FR-019, FR-020	Repeat-run test (same input, 10 runs)
NFR-REL-04	Branch failure isolation. A failure in one branch (e.g. NLP) shall not crash the pipeline. The listing shall be reported as Needs review with an explanation, not as clean.	Must	Proposed	Stub-failure test
NFR-REL-05	Leakage-free evaluation. Splits are by product ID before error injection, with duplicate checks across splits, so reported metrics are trustworthy.	Must	Planning §8.2	Split audit script
NFR-REL-06	Reproducibility. Random seeds, split ratio, class list and model versions shall be recorded so that results can be regenerated.	Must	Planning §10.2	Re-run produces the same metrics within documented tolerance
NFR-REL-07	Honest confidence. Confidence shall be calibrated (ECE reported) and described as confidence, never as proof of correctness.	Must	FR-013, OBJ-02	AC-013 and ECE in report
NFR-REL-08	No invented claims. The NLP module shall not infer attributes that are not in the text, and templates shall never add size, material or price.	Must	FR-007, FR-021	AC-007, AC-021
NFR-REL-09	Availability. No uptime target applies, because the system is a demonstration prototype with no production deployment. Stability is judged during the final demo scenarios.	Must	Constraint	Demo rehearsal checklist

4. Usability Requirements

Seller and reviewer needs are inferred, not interviewed, so each requirement here is paired with a planned walkthrough rather than a claimed result.

ID	Requirement	Priority	Basis	Verification
NFR-USA-01	The seller shall be able to submit a listing (image, title, description, category) in one screen with one primary action.	Must	FR-001–003	Scenario walkthrough
NFR-USA-02	Every warning shall show the claimed value, predicted visual value, confidence, severity, reason code and plain-language explanation.	Must	Planning §9.1, FR-016	AC-016, SC-04
NFR-USA-03	Severity and queue status shall be visually distinguishable by text label (Critical / Major / Minor, and Ready / Needs review / Critical mismatch), not by color alone.	Should	Proposed	UI inspection

ID	Requirement	Priority	Basis	Verification
NFR-USA-04	The Before/After view shall show original and suggested text side by side, with changes highlighted and the reason given.	Must	FR-022	AC-022
NFR-USA-05	Accept, Edit and Ignore shall always be visible together, and nothing shall be applied until the seller chooses.	Must	FR-023–025	AC-023–025
NFR-USA-06	Low-confidence messages shall use the plain wording “Please verify”, and shall not present an uncertain prediction as a finding.	Must	FR-014	AC-014
NFR-USA-07	The interface shall be in English. Arabic text input is supported only through the fixed lexicon, and the UI shall state this limitation. A full Arabic UI is not required.	Should	Scope: bounded Arabic	UI inspection
NFR-USA-08	The UI shall use consistent terminology (“Product Type”, “Color”, “Review Queue”) matching the three documents.	Should	Planning terminology	Terminology checklist
NFR-USA-09	Warning overload shall be controlled. At most one warning per reason code per attribute shall be displayed, with the most severe first.	Should	Planning §7.2 risk	Scenario test with multi-flag fixtures
NFR-USA-10	Warning comprehension, queue usefulness and editing behavior shall be checked through scenario walkthroughs. The results are not yet gathered.	Should	Planning §3.1, FRS §13	Planned walkthrough record

5. Maintainability Requirements

ID	Requirement	Priority	Basis	Verification
NFR-MNT-01	The system shall be modular: CV, NLP, comparison, scoring/queue, template generation and UI as separate modules with defined interfaces.	Must	Planning §9, FRS §11	Architecture review
NFR-MNT-02	All modules shall exchange data through a single Pydantic data contract.	Must	Proposal integration strategy	Schema inspection
NFR-MNT-03	Stub modules shall exist early so that components can be replaced incrementally without changing callers.	Should	Planning §9.1	Swap test: replace the stub with the real module
NFR-MNT-04	Rules and vocabularies shall be configuration, not code: the type-group table, color-family and near-shade table, English and Arabic lexicons, reason-code templates and suggestion templates.	Must	FR-007, FR-017, FR-021	Change a rule without editing logic
NFR-MNT-05	Thresholds shall be configurable (confidence, blur, brightness, resolution) and recorded with their validation-derived values.	Must	FR-006, FR-014	Config inspection
NFR-MNT-06	Optional features shall be removable without breaking the core, following the cut order Quality Score, Missing Attribute Detection, MLflow, then Docker.	Must	FRS §5.1	Disable-flag test (FR-018 and FR-012 off)
NFR-MNT-07	Traceability shall be maintained: FR to OBJ to US to AC, and every NFR to TAC, updated through the change log.	Must	FRS §14–15	Traceability audit
NFR-MNT-08	Reproducibility artifacts shall be provided: requirements file, README, notebooks, saved weights and API usage notes.	Must	Planning §10.2	Fresh-environment run

ID	Requirement	Priority	Basis	Verification
NFR-MNT-09	Code shall carry docstrings and a tests folder with fixtures matching AC-001 to AC-025.	Should	Proposed	Repository inspection

6. System Constraints and Limitations

6.1 Constraints (fixed by the proposal)

ID	Constraint	Source
CON-01	Fashion Product Images Small is the sole dataset for training, validation, testing and evaluation.	Proposal, FRS §12.1
CON-02	Exactly five articleType classes, selected after EDA (class balance, separability, availability, label quality). The final list is not yet fixed.	Planning §6.4
CON-03	Only Product Type and Color are checked. No size, material or price.	Scope
CON-04	ResNet or MobileNet transfer learning as the primary CV model, with CLIP zero-shot as baseline and fallback.	FR-004
CON-05	English rules plus a small fixed Arabic lexicon. No broad multilingual NLP.	FR-007/008
CON-06	Templates only. No LLM-generated content.	FR-021
CON-07	Seller approval is mandatory. No auto-publishing and no live marketplace integration.	FR-023–025
CON-08	FastAPI backend and Streamlit frontend.	Proposal
CON-09	One listing per request. No batch CSV.	Out-of-scope list
CON-10	Out of scope: return prediction, dynamic pricing, price validation, material/size recognition, recommendation, competitor scraping, cloud production, LoRA fine-tuning, additional domains.	Planning §6.3
CON-11	Four fixed milestones: 16 Oct, 6 Nov, 30 Nov, 4 Dec 2026.	Planning
CON-12	Split by product ID before error injection (70/15/15 is illustrative, not approved).	Planning §8.2

6.2 Limitations (honest boundaries of the results)

ID	Limitation
LIM-01	The dataset has small, low-resolution images, and color labels may be noisy or ambiguous.
LIM-02	Listings are curated or synthetic, not real marketplace submissions. Descriptions are manually authored or synthetic because the dataset has no description column.
LIM-03	Mismatch training cases are synthetically injected, so the benefit of the lightweight classifier over the rule baseline is an empirical question.
LIM-04	Arabic coverage is deliberately narrow. Dialect, Arabizi, spelling variants and stylized color names will be missed.
LIM-05	Similar types (e.g. near shoe types) and multi-color products may be confused.
LIM-06	Unseen categories cannot be classified. They are escalated via UNSUPPORTED_CATEGORY, not forced.
LIM-07	Results can show internal consistency behavior only. They cannot prove marketplace gains, fewer returns, lower cost or shorter review time.
LIM-08	Hardware, training time and weight-download access are not yet verified.
LIM-09	The manual set (100–150 listings) is built from held-out dataset images, so it is small and is not a real-world sample.

6.3 Open decisions that constrain requirements

Open items from the FRS carry over unchanged: Q-01 (Medium-only queue state), Q-02 (combined-flag precedence), IN-01 to IN-04 (empty title/description, codec errors, malformed category), the final five classes, color-family tolerance, the confidence threshold, and the score weights.

7. Technical Requirements

ID	Requirement	Priority	Basis
NFR-TEC-01	Language and stack: Python for ML, NLP and backend.	Must	Planning §7.1
NFR-TEC-02	Backend: FastAPI orchestrates parallel CV and NLP, comparison, scoring and queue.	Must	FRS §9
NFR-TEC-03	Frontend: Streamlit provides intake, findings, queue status, Before/After and the Accept / Edit / Ignore actions.	Must	FRS §9
NFR-TEC-04	Data contract: Pydantic models define the input listing and all module outputs (attributes, confidence, claims, findings, severity, reason codes, queue status, suggestion, seller decision).	Must	Planning §9.1
NFR-TEC-05	CV model: ResNet or MobileNet with pretrained weights (transfer learning) for Product Type and Color, using model-appropriate preprocessing.	Must	FR-004/005, refs [4, 5]
NFR-TEC-06	Baseline and fallback: CLIP zero-shot, evaluated independently. It is not a preprocessing step.	Must	FR-004, ref [6]
NFR-TEC-07	Color cross-check: a dominant-color check supports the visual color decision.	Must	FR-005
NFR-TEC-08	Image quality: resolution, blur and brightness checks.	Must	FR-006
NFR-TEC-09	NLP: controlled vocabulary, regex, fuzzy matching, Arabic-character script detection.	Must	FR-007/008
NFR-TEC-10	Comparison: rule table (type groups, color families) plus a lightweight classifier (logistic regression or small MLP). Rules remain authoritative.	Must	FR-009, FR-017
NFR-TEC-11	Reason codes (fixed set): TYPE_MISMATCH, COLOR_MISMATCH, LOW_CONFIDENCE, IMAGE_BLUR, MISSING_ATTRIBUTE, UNSUPPORTED_CATEGORY.	Must	FR-016
NFR-TEC-12	Queue states (fixed set): Ready to publish, Needs review, Critical mismatch.	Must	FR-020
NFR-TEC-13	Evaluation tooling: accuracy, macro-F1, confusion matrix, ECE, precision, recall, F1, false-alarm rate, catch rate, response time.	Must	OBJ-09
NFR-TEC-14	Optional: MLflow logging, Docker packaging and CLIP linear probe, only if the core stays secure.	Could	Planning §6.2
NFR-TEC-15	Environment: hardware must run training and inference locally. Verify CPU/GPU, memory, library versions and weight downloads early.	Must	Planning §7.1

8. Use Case Diagram

8.1 Actors

Actor	Type	Source
Marketplace Seller	Primary	FRS §2
Catalog Reviewer	Primary (review user)	FRS §2
Academic Supervisor	Not a system actor (reviews documents only)	FRS §2
Operations / Support	Indirect beneficiary, no system actor role	FRS §2

No administrator actor exists. Internal modules (CV, NLP, comparison engine) are part of the system boundary, not actors.

8.2 Use case diagram

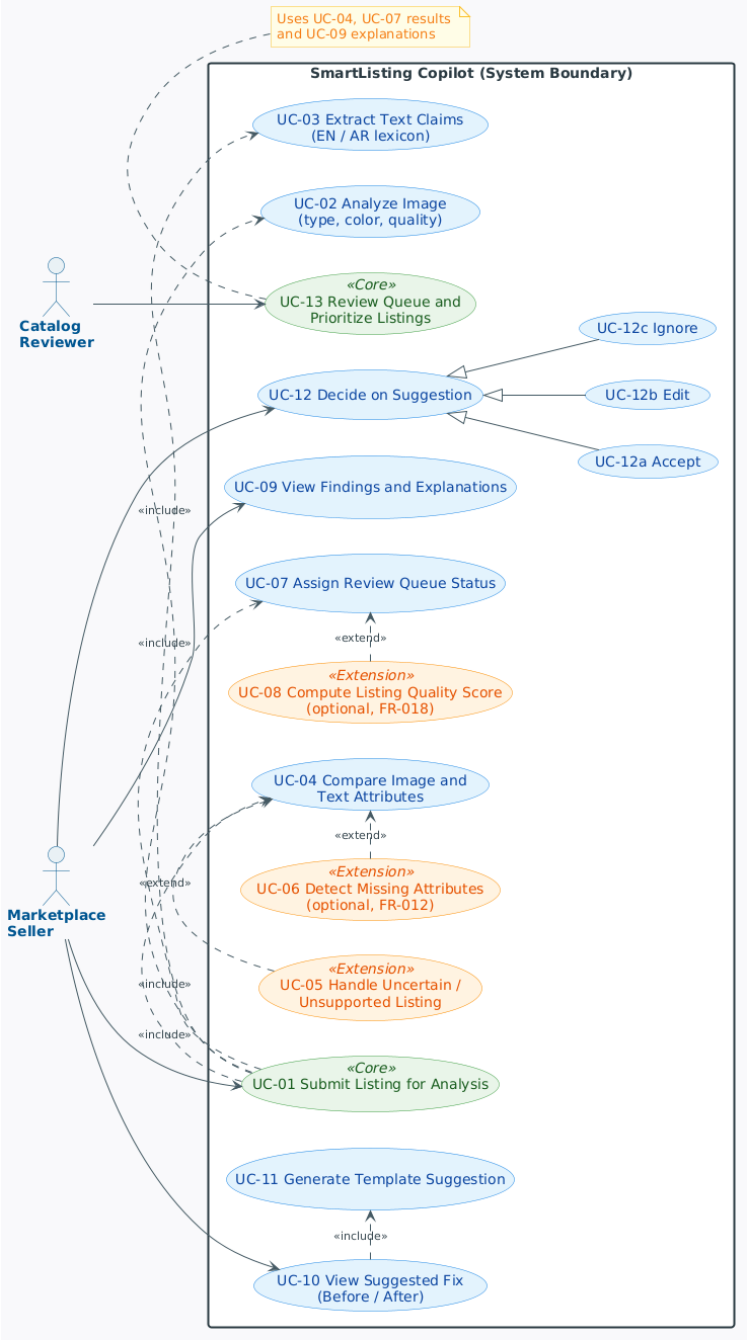


Figure 1. SmartListing Copilot use case diagram (system boundary, actors, include / extend relationships).

8.3 Actor-to-use-case matrix

Use Case	Seller	Reviewer
UC-01 Submit Listing for Analysis	✓	
UC-02 to UC-08 (internal analysis)	via UC-01	via UC-13
UC-09 View Findings and Explanations	✓	✓
UC-10 View Suggested Fix	✓	
UC-11 Generate Template Suggestion	via UC-10	

Use Case	Seller	Reviewer
UC-12 Decide on Suggestion (Accept / Edit / Ignore)	✓	
UC-13 Review Queue and Prioritize Listings		✓

9. Use Case Descriptions

UC-01 Submit Listing for Analysis

Actor	Marketplace Seller
Goal	Obtain a listing-quality assessment before publication
Related FR / OBJ / US	FR-001, 002, 003 / OBJ-01 / US-01
Preconditions	The Streamlit intake view is open and the FastAPI backend is running
Trigger	The seller presses Analyze
Main flow	1. The seller provides one image, a title, a description and a category. 2. The system validates that an image is present. 3. The system passes the image to CV, the text to NLP and the category to the scope check. 4. The system runs UC-02 and UC-03 in parallel. 5. The system continues to UC-04 and UC-07. 6. The system presents the results (UC-09).
Alternative flows	A1 Missing image: analysis does not start (confirmed behavior). A2 Unreadable or corrupt image: normal analysis cannot proceed; the error presentation is TBD (IN-03). A3 Empty title or description: reject versus treat as absent claims is TBD (IN-01/02). A4 Blank or malformed category: TBD (IN-04).
Postconditions	Findings and a queue status are available. Nothing is published.

UC-02 Analyze Image (internal)

Actor	CV module (system)
Related FR	FR-004, 005, 006, 013
Main flow	1. Preprocess the image. 2. Check resolution, blur and brightness. 3. Predict Product Type with ResNet/MobileNet (CLIP as baseline or fallback). 4. Predict Color, cross-checked with the dominant color. 5. Return predictions with confidence.
Alternative flows	A1 Blur rule triggered: IMAGE_BLUR flag. A2 Confidence below threshold: LOW_CONFIDENCE. A3 Out-of-scope image: no forced label (see UC-05).
Postconditions	Structured visual attributes, confidence and quality findings are available.

UC-03 Extract Text Claims (internal)

Actor	NLP module (system)
Related FR	FR-007, 008

Main flow	1. Detect Arabic characters. 2. Select the Arabic lexicon or the English core. 3. Normalize the text. 4. Extract only the type claim and color claim using rules and fuzzy matching.
Alternative flows	A1 No claim found: the field stays empty and nothing is inferred (feeds UC-06). A2 Mixed script: Arabic-script input routes to the Arabic lexicon, per FR-008.
Postconditions	Normalized claimed type and claimed color are available.

UC-04 Compare Image and Text Attributes (internal)

Actor	Comparison engine (system)
Related FR	FR-009, 010, 011, 013, 016, 017, 019
Preconditions	CV predictions and NLP claims exist
Main flow	1. Compare predicted and claimed type. 2. Compare predicted and claimed color (family and near-shade rules). 3. Assign severity (cross-group type = Critical; same-group type = Major; different-family color = Major; near shade = Minor or ignored). 4. Attach a reason code, confidence and an explanation. 5. Compute the Listing Inconsistency Level.
Alternative flows	A1 Consistent attributes: no finding. A2 Claim absent: see UC-06.
Postconditions	A mismatch report and an inconsistency level (Low / Medium / High) exist.

UC-05 Handle Uncertain / Unsupported Listing (extends UC-04)

Actor	Review engine (system)
Related FR	FR-014, 015
Trigger	Confidence is below threshold, or CV and CLIP disagree, or the predicted type contradicts the seller category, or CLIP favors the “other product” prompt
Main flow	1. Raise LOW_CONFIDENCE and/or UNSUPPORTED_CATEGORY. 2. Avoid a forced definitive type. 3. Route to Needs review. 4. Show “Please verify”.
Postconditions	The listing is never confidently auto-accepted.

UC-06 Detect Missing Attributes (optional, extends UC-04)

Actor	Comparison engine
Related FR	FR-012 (Should Have)
Main flow	1. A visually supported type or color is absent from the text. 2. Raise MISSING_ATTRIBUTE (type or color). 3. Route to Needs review.
Alternative flow	A1 Contradicted claim: treated as a mismatch, not as missing.
Note	Can be cut under the proposal fallback.

UC-07 Assign Review Queue Status (internal)

Actor	Review engine
Related FR	FR-020
Main flow	Clean and Low goes to Ready to publish. Any review flag (low confidence, blur, missing attribute, unsupported category) goes to Needs review. High inconsistency or any Critical goes to Critical mismatch.
Open issues	Q-01: the queue state for exactly one Major (Medium level) is undecided. Q-02: precedence among conflicting flags is undecided. No default is assumed.

UC-08 Compute Listing Quality Score (optional)

Actor	Scoring module
Related FR	FR-018 (Should Have)
Main flow	Combine image quality, attribute completeness and image-text consistency into a 0–100 heuristic. Confidence is excluded.
Open issue	The weights are TBD.

UC-09 View Findings and Explanations

Actors	Marketplace Seller, Catalog Reviewer
Related FR / OBJ / US	FR-013, 016, 017 / OBJ-05 / US-02, US-04
Main flow	The actor opens the results. For each warning the system shows the claimed value, predicted value, confidence, severity, reason code and plain-language rationale.
Alternative flow	A1 No findings: the system shows the status “Ready to publish” and the supporting predictions.
Postcondition	Evidence is visible. Confidence is not presented as correctness.

UC-10 View Suggested Fix (Before / After)

Actor	Marketplace Seller
Related FR / OBJ / US	FR-021, 022 / OBJ-07 / US-07
Preconditions	At least one finding supports a correction
Main flow	1. The system generates a template suggestion (UC-11). 2. The system shows the original and the suggested text side by side, with changes highlighted and reasons stated.
Alternative flow	A1 Uncertain or unsupported listing: the suggestion uses only model-supported attributes, or the system shows no suggestion for an unsupported type.
Postcondition	The seller can compare before deciding. Nothing is applied.

UC-11 Generate Template Suggestion (internal)

Actor	Generation module
Related FR	FR-021
Main flow	Fill controlled templates using only predicted Product Type and Color. No LLM, no size, no material, no price.

UC-12 Decide on Suggestion (Accept / Edit / Ignore)

Actor	Marketplace Seller
Related FR / OBJ / US	FR-023, 024, 025 / OBJ-08 / US-08
Preconditions	A suggestion is displayed
Main flow	The seller chooses one option. 12a Accept: the correction is recorded as seller-approved in the current review flow. 12b Edit: the seller changes the wording and confirms, and the revised wording is retained. 12c Ignore: the original text is kept and the suggestion is left unapplied.
Alternative flow	A1 No decision: nothing is applied.
Postcondition	No external publish action is performed in any branch. The duration and representation of seller decisions are an open item.

UC-13 Review Queue and Prioritize Listings

Actor	Catalog Reviewer
Related FR / OBJ / US	FR-016, 017, 019, 020 / OBJ-06 / US-04, US-05, US-06
Main flow	1. The reviewer opens the queue. 2. Listings appear grouped as Critical mismatch, Needs review and Ready to publish. 3. The reviewer opens a flagged listing and reads its reason codes and evidence.
Note	The MVP defines no reviewer accounts and no separate reviewer workflow beyond viewing the queue and findings. The queue is a local decision-support label only.

10. Acceptance Criteria (Non-Functional and Use-Case)

AC-001 to AC-025 stay as defined in the FRS. The criteria below are additional (TAC-xx). They are planned fixture-driven tests, not executed results. Numeric thresholds stay TBD until frozen.

10.1 Non-functional acceptance criteria

ID	Linked NFR	Given / When / Then
TAC-01	PERF-01, 02	Given a supported listing; when the pipeline runs; then a timing log reports CV, NLP, comparison, queue and total time, and the total is no more than the slowest branch plus downstream stages.
TAC-02	PERF-03	Given the prototype timing results; when the final evaluation begins; then a dated response-time target is documented.
TAC-03	PERF-04, 05, 06	Given held-out data; when evaluation runs; then the report contains the required metrics per attribute, per mismatch type and per language, with thresholds frozen beforehand.
TAC-04	PERF-07, REL-05	Given synthetic and manual results; when the report is produced; then they appear in separate tables and no product ID occurs in more than one split.
TAC-05	SEC-01	Given each of Accept, Edit and Ignore; when selected; then no outbound publish call is made (verified by interaction test and code search).
TAC-06	SEC-02, 03	Given valid, corrupt and wrong-type files and special-character text; when submitted; then the system handles them without crash or code execution and shows a non-technical error.
TAC-07	SEC-04	Given a forced internal error; when it occurs; then no stack trace or file path appears in the UI.
TAC-08	SEC-06, 07	Given a network monitor; when a listing is analyzed; then no uploaded content leaves the local environment.

ID	Linked NFR	Given / When / Then
TAC-09	REL-01, 02	Given a blurred image, a low-confidence image and an out-of-scope image; when analyzed; then each routes to Needs review and none is Ready to publish. The primary model disabled leads to fallback or escalation.
TAC-10	REL-03	Given identical input run 10 times; when compared; then severity, level, queue status and suggestion are identical.
TAC-11	REL-04	Given a simulated NLP failure; when the pipeline runs; then the listing is routed to Needs review with an explanation and the application keeps running.
TAC-12	REL-06, MNT-08	Given a fresh environment and the saved seeds and weights; when the README steps are followed; then reported metrics are reproduced within the documented tolerance.
TAC-13	USA-01, 04, 05	Given a scenario walkthrough; when a participant submits, reads a warning, and chooses Accept / Edit / Ignore; then the steps are completable and results are recorded (not yet conducted).
TAC-14	USA-02, 06	Given each reason code; when displayed; then claimed value, predicted value, confidence, severity and rationale are all present and “Please verify” appears for low confidence.
TAC-15	USA-09	Given a multi-flag fixture; when displayed; then warnings are de-duplicated and ordered by severity.
TAC-16	MNT-04, 05	Given a changed color-family table or confidence threshold in configuration; when the system restarts; then behavior changes without editing source logic.
TAC-17	MNT-06	Given FR-018 and FR-012 disabled; when the core workflow runs; then FR-006 findings, reason codes, queue, suggestions and seller control still work.
TAC-18	MNT-03	Given a stub replaced with its real module; when the integrated flow runs; then no caller code changes.

10.2 Use-case acceptance criteria

ID	Use Case	Given / When / Then
TAC-UC01	UC-01	Given a complete valid submission; when Analyze is pressed; then the image, text and category reach their branches and results are shown. Given no image; then analysis does not start.
TAC-UC04	UC-04	Given a cross-group type contradiction and a different-family color contradiction; when compared; then the type finding is Critical and the color finding is Major, each with a reason code and explanation.
TAC-UC05	UC-05	Given CV and CLIP disagreement; when the scope check runs; then UNSUPPORTED_CATEGORY is raised, no forced type is shown, and status is Needs review.
TAC-UC07	UC-07	Given clean, low-confidence and Critical fixtures; when routed; then the statuses are Ready, Needs review and Critical mismatch. Medium-only and combined-flag cases are blocked pending Q-01/Q-02.
TAC-UC09	UC-09	Given any warning; when the actor views findings; then it shows a defined reason code, evidence and rationale.
TAC-UC10	UC-10, 11	Given a mismatched listing; when the seller opens the suggestion; then only predicted Type/Color appear, no size/material/price, and changes are highlighted.
TAC-UC12	UC-12	Given a displayed suggestion; when the seller picks Accept, Edit or Ignore; then the choice is respected, Ignore keeps the original text, and nothing is published.
TAC-UC13	UC-13	Given listings in all three states; when the reviewer opens the queue; then they are grouped by state and each opens to its reason codes.

11. Requirements Traceability Matrix

11.1 Non-functional requirements to objectives, sources, use cases and tests

NFR ID	Requirement (short)	Project Objective	Source	Related FR / UC	Acceptance
NFR-PERF-01	Measure end-to-end response time	OBJ-09	SC-10	UC-01	TAC-01
NFR-PERF-02	Parallel CV and NLP	OBJ-01	FRS §9	UC-02, UC-03	TAC-01
NFR-PERF-03	Freeze response-time target	OBJ-09	Proposal	UC-01	TAC-02
NFR-PERF-04	Report CV accuracy / F1 / ECE	OBJ-02	SC-02	FR-004, 005 / UC-02	TAC-03
NFR-PERF-05	Report mismatch P / R / F1 / false alarms	OBJ-04	SC-03	FR-009–011 / UC-04	TAC-03
NFR-PERF-06	Report NLP P / R / F1 by language	OBJ-03	OBJ-03	FR-007, 008 / UC-03	TAC-03
NFR-PERF-07	Separate synthetic vs manual	OBJ-09	SC-09	All	TAC-04
NFR-PERF-08	Load models once	OBJ-01	Proposed	UC-02	TAC-01
NFR-PERF-09	One listing per request	OBJ-01	Out-of-scope	UC-01	Scope review
NFR-SEC-01	No autonomous publishing	OBJ-08	SC-07	FR-023–025 / UC-12	TAC-05, AC-023–025
NFR-SEC-02	Image upload validation	OBJ-01	IN-03	FR-001 / UC-01	TAC-06
NFR-SEC-03	Text and category validation	OBJ-01	Pydantic	FR-002, 003 / UC-01	TAC-06
NFR-SEC-04	Safe error responses	OBJ-01	Proposed	UC-01	TAC-07
NFR-SEC-05	Data provenance	OBJ-09	Planning §8	UC-01	Data-card check
NFR-SEC-06	Data minimization / local inference	OBJ-01	Proposed	UC-02	TAC-08
NFR-SEC-07	Local-only deployment	OBJ-01	Scope	All	TAC-08
NFR-SEC-08	Dependency hygiene	OBJ-09	Proposed	All	Requirements-file check
NFR-SEC-09	No auth layer implied	OBJ-01	FRS §2	All	Scope review
NFR-REL-01	Degrade to Needs review	OBJ-06	SC-08	FR-014, 015 / UC-05	TAC-09, AC-014, AC-015
NFR-REL-02	CLIP fallback	OBJ-02	FR-004	UC-02, UC-05	TAC-09
NFR-REL-03	Deterministic decisions	OBJ-06	FR-017, 019, 020	UC-04, UC-07	TAC-10
NFR-REL-04	Branch failure isolation	OBJ-01	Proposed	UC-01	TAC-11
NFR-REL-05	Leakage-free splits	OBJ-09	Planning §8.2	None	TAC-04
NFR-REL-06	Reproducibility	OBJ-09	Planning §10.2	None	TAC-12
NFR-REL-07	Honest confidence	OBJ-02, OBJ-05	FR-013	UC-09	TAC-03, AC-013
NFR-REL-08	No invented claims	OBJ-03, OBJ-07	FR-007, 021	UC-03, UC-11	AC-007, AC-021
NFR-REL-09	No uptime target	OBJ-01	Constraint	All	Demo checklist
NFR-USA-01	One-screen submission	OBJ-01	FR-001–003	UC-01	TAC-13
NFR-USA-02	Full evidence per warning	OBJ-05	FR-016	UC-09	TAC-14, AC-016
NFR-USA-03	Text labels, not color alone	OBJ-05	Proposed	UC-09	UI inspection
NFR-USA-04	Before/After side by side	OBJ-07	FR-022	UC-10	AC-022
NFR-USA-05	Accept / Edit / Ignore visible	OBJ-08	FR-023–025	UC-12	TAC-13, AC-023–025

NFR ID	Requirement (short)	Project Objective	Source	Related FR / UC	Acceptance
NFR-USA-06	"Please verify" wording	OBJ-05, OBJ-06	FR-014	UC-05	TAC-14, AC-014
NFR-USA-07	English UI, Arabic limitation stated	OBJ-03	Scope	UC-03	UI inspection
NFR-USA-08	Consistent terminology	OBJ-05	Planning	All	Terminology checklist
NFR-USA-09	Control warning overload	OBJ-06	Planning §7.2	UC-09	TAC-15
NFR-USA-10	Scenario walkthroughs	OBJ-09	Planning §3.1	UC-09, UC-12	TAC-13
NFR-MNT-01	Modular design	OBJ-01	Planning §9	All	Architecture review
NFR-MNT-02	Single Pydantic contract	OBJ-01	Proposal	All	Schema inspection
NFR-MNT-03	Replaceable stubs	OBJ-01	Planning §9.1	All	TAC-18
NFR-MNT-04	Rules as configuration	OBJ-04	FR-017, 021	UC-04, UC-11	TAC-16
NFR-MNT-05	Configurable thresholds	OBJ-06	FR-006, 014	UC-02, UC-05	TAC-16
NFR-MNT-06	Removable optional features	OBJ-01	FRS §5.1	UC-06, UC-08	TAC-17
NFR-MNT-07	Maintained traceability	OBJ-09	FRS §14–15	All	Traceability audit
NFR-MNT-08	Reproducibility artifacts	OBJ-09	Planning §10.2	None	TAC-12
NFR-MNT-09	Docstrings and tests	OBJ-09	Proposed	All	Repository inspection

11.2 Use cases to functional requirements, objectives and stories

Use Case	Related FRs	Objective	User Story	Acceptance
UC-01 Submit Listing	FR-001, 002, 003	OBJ-01	US-01	AC-001–003, TAC-UC01
UC-02 Analyze Image	FR-004, 005, 006, 013	OBJ-02, OBJ-01	US-02	AC-004–006, AC-013
UC-03 Extract Text Claims	FR-007, 008	OBJ-03	US-03	AC-007, AC-008
UC-04 Compare Attributes	FR-009, 010, 011, 016, 017, 019	OBJ-04, OBJ-05	US-04, US-06	AC-009–011, 016, 017, 019, TAC-UC04
UC-05 Uncertain / Unsupported	FR-014, 015	OBJ-05, OBJ-06	US-05	AC-014, AC-015, TAC-UC05
UC-06 Missing Attributes	FR-012	OBJ-06	US-05	AC-012
UC-07 Assign Queue Status	FR-020	OBJ-06	US-06	AC-020, TAC-UC07
UC-08 Quality Score	FR-018	OBJ-01	US-06	AC-018
UC-09 View Findings	FR-013, 016, 017	OBJ-05	US-02, US-04	AC-013, AC-016, TAC-UC09
UC-10 View Suggested Fix	FR-021, 022	OBJ-07	US-07	AC-021, AC-022, TAC-UC10
UC-11 Generate Template	FR-021	OBJ-07	US-07	AC-021
UC-12 Decide on Suggestion	FR-023, 024, 025	OBJ-08	US-08	AC-023–025, TAC-UC12
UC-13 Review Queue	FR-016, 017, 019, 020	OBJ-06	US-04, US-05, US-06	AC-016, 017, 019, 020, TAC-UC13

11.3 Objective coverage check

Objective	Functional coverage	Non-functional / use-case coverage
OBJ-01	FR-001–003, 006, 018	UC-01, NFR-PERF-02, SEC-02/03, REL-04, MNT-01–03

Objective	Functional coverage	Non-functional / use-case coverage
OBJ-02	FR-004, 005	UC-02, NFR-PERF-04, REL-02, REL-07
OBJ-03	FR-007, 008	UC-03, NFR-PERF-06, USA-07
OBJ-04	FR-009–011	UC-04, NFR-PERF-05, MNT-04
OBJ-05	FR-013, 016, 017	UC-09, NFR-USA-02/03/06
OBJ-06	FR-012, 014, 015, 019, 020	UC-05–07, UC-13, NFR-REL-01/03, USA-09
OBJ-07	FR-021, 022	UC-10, UC-11, NFR-USA-04, REL-08
OBJ-08	FR-023–025	UC-12, NFR-SEC-01, USA-05
OBJ-09	None (as in the FRS, evaluation is not a feature)	NFR-PERF-01/03/07, REL-05/06, MNT-07/08

12. Open Items and Notes

No.	Item
1	Numbers are deliberately absent. The proposal forbids setting thresholds before baseline experiments. If the supervisor requires a target (for example a latency limit) to be written in now, it is added as a decision in the change log, with a date.
2	Q-01, Q-02 and IN-01 to IN-04 are inherited from the FRS (Member 5). TAC-UC07 stays partially blocked until the team records a decision.
3	Identifier scheme. NFR-xx, UC-xx, TAC-xx and CON/LIM are defined by this document. If the supervisor's template expects a different scheme, the identifiers are renamed and section 11 is updated accordingly.
4	Sign-off. Reviewer and approval fields are pending (see section 17.5).

13. Planning the Approach

The table below shows how the steps of the Requirements Gathering Process Guide are covered by this part of the documentation.

Guide step	Already covered in earlier documents?	Section in this document
2. Plan the Approach	Partly (milestones only)	Section 13
3. Conduct Elicitation	Partly (FRS §3, documentary only)	Section 14
4. Document Observations and Findings	Partly (no central register)	Section 15
5. Analyze and Prioritize	Partly (MoSCoW tags, no analysis)	Section 16
6. Verify and Validate	Partly (FRS §13 covers FRs only)	Section 17
7. Manage Changes	Partly (FRS §15 covers FRs only)	Section 18

13.1 Objectives of this requirements activity

Establish a verified, prioritized, traceable set of non-functional requirements and use cases for the SmartListing Copilot proof of concept. It must be consistent with FR-001 to FR-025 and OBJ-01 to OBJ-09.

13.2 High-level scope

In scope for this activity: NFR-PERF, SEC, REL, USA, MNT and TEC; CON-01 to CON-12; LIM-01 to LIM-09; UC-01 to UC-13; TAC criteria; and traceability. Out of scope: any new functional feature, and everything in the out-of-scope list (Planning §6.3).

13.3 Timeline

Milestone	Date	Requirements activity
Planning / literature / requirements	16 Oct 2026	Documents submitted. Elicitation is documentary.
Analysis and design	6 Nov 2026	Resolve open decisions (Section 15.3). Run walkthroughs. Freeze the requirements baseline (Section 18.1).
Implementation	30 Nov 2026	Changes only through change control.
Testing / final presentation	4 Dec 2026	Execute TAC and AC fixtures. Report results.

Detailed scheduling belongs to Member 2, so only the dependency points are shown here.

13.4 Required resources

Resource	Need	Status
People	Six team members (requirements owners are Members 5 and 6), academic supervisor (if applicable)	Supervisor role unconfirmed
Participants for elicitation	Seller proxies and reviewer proxies (e.g. classmates acting as sellers)	Not yet identified
Materials	The two submitted PDFs, the Revised Scope proposal, a Streamlit stub as a rough prototype	Prototype not built
Tools	A shared tracker for the register (Section 15), a change log	Choice TBD
Time	Walkthrough sessions between 16 Oct and 6 Nov	Not scheduled

13.5 Elicitation techniques, tailored to audience

Technique	Audience	Used for	Status
One-on-one interview	Academic supervisor	Scope confirmation, Q-01, Q-02, IN-01 to IN-04, sign-off authority	Planned
Collaborative workshop	The six team members	Freezing thresholds, class groups, color-family tolerance, and cut order	Planned
User survey	Seller proxies (short form, 5–8 questions)	Warning comprehension (NFR-USA-02, 06), terminology (USA-08)	Planned
Interface shadowing / think-aloud	Seller and reviewer proxies using the Streamlit stub	One-screen submission, overload, queue usefulness (USA-01, 09, 10)	Planned, needs prototype
Document analysis	Existing PDFs	Already done	Completed

> **Example: choosing a technique per open item.** IN-01 (empty title) is a policy question, so it goes to the supervisor interview. NFR-USA-09 (warning overload) is a behavior question that cannot be settled by asking, so it goes to shadowing with the stub.

14. Conducting Elicitation

14.1 Principle

Elicit underlying business needs and problems first, and tools or design details later. The Revised Scope fixes FastAPI, Streamlit and ResNet/MobileNet as constraints. They are not needs and are not used to justify requirements.

14.2 Need statements (inferred, not interviewed)

ID	Underlying need	Source	Evidence status	Traced to
N-01	Sellers need to learn about image-text contradictions before publication	Planning §2–3	Documentary, inferred	UC-01, UC-04
N-02	Reviewers need to know where to look first	Planning §3	Inferred	UC-13, NFR-USA-09
N-03	Users need to understand why something was flagged	Planning §3	Inferred	UC-09, NFR-USA-02
N-04	Sellers need to keep control of their own wording	Planning §3	Inferred	UC-12, NFR-SEC-01
N-05	Everyone needs uncertain cases to be called uncertain, not guessed	Planning §3.1	Inferred	UC-05, NFR-REL-01
N-06	The team needs results that can be reproduced and defended academically	Planning §3.1	Inferred	NFR-REL-05, 06

14.3 Separating needs from solutions

Statement as it might arrive	Solution-focused?	Underlying need to record
“Use a neural network for color”	Yes	Color claims in the listing must be checked against the image (N-01)
“Show a red banner for mismatches”	Yes	Severity must be noticeable without relying on color alone (NFR-USA-03)
“Make it fast”	No, but vague	Measured response time, with a target frozen after the first timing (NFR-PERF-01, 03)

14.4 Planned elicitation questions for the sessions

Audience	Questions
Supervisor	Is a Medium-only listing “Needs review” or “Ready to publish”? Should an empty title be rejected? Who signs off the baseline?
Seller proxies	Can you tell what each warning means? Would you trust the suggestion before editing it?
Reviewer proxies	Does the queue order help you decide what to open first?

No answers exist yet. They are recorded in the register (Section 15) when sessions occur.

15. Documenting Observations and Findings

15.1 Central register

The register is a single table (spreadsheet or tracker) with these fields:

Field	Meaning
Item ID	FR-xxx, NFR-xxx, CON-xx, LIM-xx, UC-xx, Q-xx, IN-xx, N-xx
Category	Functional / Non-functional / Constraint / Limitation / Open decision / Need
Statement	The requirement or observation
Source	Document, session, participant
Elicitation method	Document analysis, interview, workshop, survey, shadowing
Priority	MoSCoW (Section 16)
Status	Draft / Reviewed / Validated / Baselined / Changed

Field	Meaning
Owner	Member or role (TBD)

15.2 Categorization of everything recorded so far

Category	Items	Count
Functional requirements (what it must do)	FR-001 to FR-025	25
Non-functional requirements (how it must perform)	NFR-PERF (9), SEC (9), REL (9), USA (10), MNT (9), TEC (15)	61
Absolute project constraints	CON-01 to CON-12	12
Limitations (boundaries on results)	LIM-01 to LIM-09	9
Use cases	UC-01 to UC-13 (UC-12 has branches 12a–c)	13
Open decisions	Q-01, Q-02, IN-01 to IN-04	6
Inferred needs	N-01 to N-06	6

15.3 Open-decision register

ID	Question	Technique (13.5)	Owner	Needed by	Status
Q-01	Queue state for one Major (Medium level)	Supervisor interview	TBD	6 Nov	Open
Q-02	Precedence among conflicting flags	Workshop	TBD	6 Nov	Open
IN-01	Empty title: reject or treat as absent	Supervisor interview	TBD	6 Nov	Open
IN-02	Empty description: reject or treat as absent	Supervisor interview	TBD	6 Nov	Open
IN-03	Unreadable or corrupt image handling	Workshop	TBD	6 Nov	Open
IN-04	Blank or malformed category handling	Workshop	TBD	6 Nov	Open
D-01	Final five EDA classes and class groups	EDA, then workshop	TBD	6 Nov	Open
D-02	Color-family and near-shade tolerance	Workshop	TBD	6 Nov	Open
D-03	Confidence, blur and brightness thresholds	Validation data	TBD	Before held-out testing	Open
D-04	Response-time target (NFR-PERF-03)	After first timing	TBD	Before final evaluation	Open
D-05	Quality Score weights	Workshop	TBD	6 Nov	Open

16. Analyzing and Prioritizing

16.1 Feasibility, value and dependency assessment (by category)

Category	Technical feasibility	Business / project value	Key dependencies
Performance	Feasible. Measurement is easy and targets are deferred.	High: OBJ-09 and SC-10 require the measurements.	Working integrated pipeline, timing hooks
Security	Feasible. Scope is small (local prototype, no accounts).	High for NFR-SEC-01 (core promise). Moderate for the rest.	Pydantic contract, upload handling
Reliability	Feasible with caveats. Calibration and fallbacks need models first.	High: uncertainty handling is the core of the product idea.	Trained model, CLIP, thresholds (D-03)

Category	Technical feasibility	Business / project value	Key dependencies
Usability	Feasible. The risk is that findings are unconfirmed without walkthroughs.	High for explanations. Moderate for polish items.	Streamlit stub, walkthrough participants
Maintainability	Feasible. Mostly design discipline.	Moderate for the team. High for traceability.	Config files, module interfaces
Technical	Feasible, per Planning §7.1, but hardware is unverified.	Required by the proposal.	Compute check (NFR-TEC-15)

16.2 Dependency view

Requirement	Depends on	Why
NFR-PERF-03	First prototype timing	Target can't be set before measurement
NFR-PERF-04	D-01, D-03	Metrics need final classes and frozen thresholds
NFR-REL-01	D-03	Low-confidence routing needs the threshold
NFR-REL-02	CLIP baseline	Fallback must exist
NFR-USA-10	Streamlit stub	Nothing to walk through otherwise
NFR-MNT-06	FR-012, FR-018 flags	Cut order must be switchable
UC-07 / TAC-UC07	Q-01, Q-02	Routing edge cases undefined

16.3 MoSCoW prioritization of the non-functional requirements

Priority	Requirements	Count
Must have	PERF-01 to 07, 09; SEC-01, 02, 03, 05, 07, 09; REL-01 to 09; USA-01, 02, 04, 05, 06; MNT-01, 02, 04, 05, 06, 07, 08; TEC-01 to 13, 15	48
Should have	PERF-08; SEC-04, 06, 08; USA-03, 07, 08, 09, 10; MNT-03, 09	12
Could have	TEC-14 (MLflow, Docker, CLIP linear probe)	1
Won't have	Uptime or availability targets, throughput or concurrency targets, authentication / authorization, production cloud deployment, batch processing	n/a

Use cases. Must have: UC-01 to 05, 07, 09 to 13. Should have: UC-06 (FR-012) and UC-08 (FR-018), matching the FRS cut order (Quality Score, Missing Attribute, MLflow, Docker).

Reconciliation. NFR-MNT-06 is a Must even though the features it protects are Should, because the ability to cut them safely is what keeps the core deliverable secure.

16.4 Kano classification (proposed hypothesis)

No survey was run, so this is the team's expectation, to be checked in the survey (Section 13.5).

Kano class	Examples	Reasoning
Basic (must-be)	NFR-SEC-01 (no auto-publish), REL-01 (uncertain goes to review), USA-05 (Accept / Edit / Ignore)	Absence would break trust, and presence earns no praise
Performance (more is better)	PERF-04/05 (accuracy, false-alarm rate), PERF-01 (speed)	Satisfaction scales with the measured level
Delighter	USA-04 (highlighted Before/After), USA-02 (full evidence per warning), FR-018 (Quality Score)	Not expected, but valued if present
Indifferent	MNT-09 (docstrings), SEC-08 (dependency pinning)	Invisible to sellers and reviewers, though they matter to the team

17. Verification and Validation

17.1 Definitions (as in FRS §13.1)

Verification checks the document against sources, completeness, consistency and testability. Validation checks that the requirements match what real stakeholders need.

17.2 Status of this document

Activity	Status
Cross-document consistency review (against the Planning Document and FRS)	Done during drafting
Traceability inspection (NFR to OBJ to UC to TAC)	Done during drafting
Testability inspection of TAC	Done during drafting
Scenario walkthroughs with stakeholders	Planned, not conducted
Wireframe or prototype review	Planned, needs the Streamlit stub
Formal recorded sign-off	Pending

17.3 Review criteria summary for NFR and UC items

Category	Correctness	Completeness	Consistency	Feasibility	Testability	Scope alignment	Open issue
Performance	Checked	Open	Checked	Checked	Checked	Checked	D-04 target pending
Security	Checked	Checked	Checked	Checked	Checked	Checked	IN-03 limits pending
Reliability	Checked	Checked	Checked	Checked	Checked	Checked	D-03 threshold pending
Usability	Checked	Open	Checked	Checked	Checked	Checked	Walkthrough results pending
Maintainability	Checked	Checked	Checked	Checked	Checked	Checked	None structural
Technical	Checked	Checked	Checked	Open	Checked	Checked	Hardware unverified
Use cases	Checked	Open	Checked	Checked	Checked	Checked	Q-01, Q-02, IN-01 to IN-04

"Checked" means document-level inspection only. It is not stakeholder validation.

17.4 Planned walkthrough scenarios

ID	Scenario	Participants	Validates	Prototype needed
WT-01	Seller submits a clean listing	Seller proxy	UC-01, NFR-USA-01	Yes
WT-02	Seller submits a cross-group type mismatch and reads the warning	Seller proxy	UC-04, 09, NFR-USA-02, 06	Yes
WT-03	Seller sees a Before/After suggestion and chooses Accept, then Edit, then Ignore	Seller proxy	UC-10, 12, NFR-USA-04, 05	Yes
WT-04	Reviewer opens a queue with mixed states	Reviewer proxy	UC-13, NFR-USA-09	Yes
WT-05	Blurred or out-of-scope image submitted	Both	UC-05, NFR-REL-01	Yes
WT-06	Open-decision walkthrough (Q-01, Q-02, IN-01 to IN-04)	Supervisor	Section 15.3	No

17.5 Sign-off record

Primary owners	Pending: the supervisor and team lead need to be designated
Reviewer	Pending: no named reviewer
Review date	Not yet conducted
Scope of review	Sections 0–18, NFR, UC, TAC, RTM
Decision	Pending: no approval or rejection recorded
Open issues	Section 15.3 (all 11 items)
Condition	Sign-off is to be recorded before active implementation begins, with the date set by the team

18. Managing Changes

18.1 Baseline configuration

Baseline	Contents	Status
BL-0 (submission)	Planning Document, FRS, and this NFR and Use Case document as submitted on 16 Oct 2026	Submitted, not approved
BL-1 (proposed)	BL-0 plus resolved open decisions (Section 15.3), frozen at the analysis-and-design milestone	Target: 6 Nov 2026

18.2 Traceability maintained through implementation

The traceability matrix (Section 11) is kept current. Every change must update the FR → OBJ → US → UC → AC / TAC chain, and every NFR keeps its link to an objective and a test.

18.3 Change-control process

Step	Action
1	Raise a change request: ID, description, reason, requester
2	Classify it: clarification, correction, new item, removal, or scope change
3	Assess impact: affected FR / NFR / UC / AC / TAC, schedule, the four milestones, and the cut order
4	Scope check: a request touching the out-of-scope list (Planning §6.3) is rejected by default unless the supervisor explicitly approves a scope change
5	Decide: team, plus supervisor for scope or priority changes
6	Update the affected documents and the register
7	Record it in the change log with evidence, decider and date
8	Re-baseline at the next milestone

18.4 Handling the situations named in the guide

Situation	Handling
Evolving scope	Goes through steps 1–8. Anything outside the five classes, two attributes or single dataset is a scope change.
Mid-project discovery (e.g. EDA shows a class is unusable)	Raise a change request with the EDA evidence and re-select the class (D-01). Update UC, TAC and metrics.
Missed item	Add it to the register as Draft and assess it under step 3. Do not add it silently.

Situation	Handling
Open decision resolved	Record the decision, update FR-020 / AC-020, TAC-UC07 and the routing table together (FRS §12.3)

18.5 Change log (this document)

Change ID	Section	Previous issue	Revision made	Reason	Approval
CH-N01	13–18	Process steps 2–7 of the Requirements Gathering Process Guide not reflected	Added planning, elicitation, register, prioritization, V&V and change management sections	Align with the guide	Pending
CH-N02	15.3	Open items scattered across documents	Consolidated Q-01, Q-02, IN-01 to IN-04 and D-01 to D-05	Single source of truth	Pending
CH-N03	16	NFR priorities not recorded	Added MoSCoW and Kano analysis	Prioritization step	Pending